Benchmarking cell segmentation for subcellular spatial transcriptomics

Pengfei Guan[1*], Yijia Qiu[2*], Yiming Luo[1], Yuhang Yang[2], Kai Zhang[2, ✉], Yang Chen[1, ✉], Enhong Chen[2, ✉], and Lin Shen[1, ✉]

[1]Department of Gastrointestinal Oncology, Key Laboratory of Carcinogenesis and Translational Research (Ministry of Education), Peking University Cancer Hospital and Institute, Beijing, China
[2]State Key Laboratory of Cognitive Intelligence, University of Science and Technology of China, Hefei, China
[*]These authors contributed equally to this work
[✉]kkzhang08@ustc.edu.cn, yang_chen@bjcancer.org.cn, cheneh@ustc.edu.cn, and shenlin@bjmu.edu.cn

ABSTRACT

Tracing molecular features and spatial relationships at the single-cell level is an essential step in spatial omics research, and cell segmentation, which delineates individual cell boundaries from histology, nuclear staining, and spatial transcriptomics data, provides its foundation. Accurate segmentation underpins a broad range of downstream analyses, from quantifying cell morphology, phenotyping specific cell types to dissecting tissue microenvironments and cellular interactions. Conventional methods segment cells from nuclear or membrane staining alone, whereas newer methods also use the positions of transcripts. And some methods refine an existing segmentation rather than starting from scratch. These methods differ widely in their inputs, algorithms, and computational demands, and each has typically been evaluated on its own datasets and metrics. As a result, a systematic benchmark to guide the choice among them has been lacking. Here we evaluate 16 cell segmentation methods on 31 datasets from five platforms, comprising 445 completed runs of the 466 applicable tasks. We assess their accuracy at the mask and transcript levels, their robustness and computational efficiency across platforms and tissues. We further test their behaviour in three challenging scenarios, namely whole-slide inputs, gene panels ranging from a few hundred genes to the whole transcriptome, and tissues ranging from sparse to dense, and quantify how segmentation errors propagate into downstream analysis using protein-defined cell identities. Our results indicate that no method was both highly accurate and broadly applicable, so the choice of a segmentation method and strategy should be tailored to the characteristics of the data and the requirements of downstream analysis. To support this choice, we provide a compact set of recommended metrics, a step-by-step workflow, and an extensible recommendation tool that suggests suitable methods from the characteristics of the data, availability of hardware, and requirements of downstream analysis in any practical situation.

1 Introduction

Spatial transcriptomics (ST) technologies include sequencing-based methods such as Visium, Slide-seq[1], and Stereo-seq[2], as well as imaging-based methods such as MERFISH[3], STARmap[4], and Xenium[5–8]. These technologies capture gene expression profiles together with their spatial coordinates, enabling in situ characterization of gene expression within the native tissue environment[9,10]. This allows researchers to move beyond conventional cell clustering to identify fine tissue structures and cellular interactions, with applications ranging from mapping tissue anatomy and spatial continuity to detecting domain-specific marker genes.[11–18].

Delineating precise cell boundaries and assigning transcripts to individual cells is a fundamental step for any analysis of ST data. Image-based segmentation methods, such as Cellpose, Mesmer, and approaches based on watershed pipelines, star-convex polygon models, and other deep learning architectures, are still widely adopted because they are mature and computationally efficient, which makes them attractive for large ST datasets. However, because they lack direct transcriptomic evidence for refining boundaries, their accuracy often suffers when membrane signals are weak or cells are densely packed[19,20]. In recent years, a range of transcript-informed algorithms has been developed, from statistical point-cloud models and graph-based models built on community detection or graph neural networks to image-guided multimodal methods that combine staining with transcript information[19–21]. These methods have steadily improved the accuracy of cell boundaries, but they are limited by processing throughput and data sparsity.

More critically, it remains unclear whether these methods generalize across ST platforms and tissue types, how their transcript assignments can be compared quantitatively, whether they resolve contamination from neighbouring cells, and how their errors affect downstream analyses[22]. Answering these questions is difficult, because the methods differ in their inputs and outputs, the platforms differ in imaging and data formats, and independent reference annotations are scarce. As subcellular ST is rapidly adopted and segmentation errors propagate into every downstream analysis, a systematic benchmark that addresses these difficulties is urgently needed.

To address these challenges, we developed SpaSegBench, a comprehensive benchmarking framework for cell segmentation in ST. We systematically evaluated 16 cell segmentation methods on 31 datasets from 7 technologies of 5 platforms, focusing on technologies that resolve transcript positions at subcellular resolution. We categorized the methods according to their underlying algorithms and dependence on existing segmentation priors, and assessed their accuracy, downstream impact, robustness, and computational efficiency using standardized evaluation procedures. We further examined their performance in challenging scenarios, including the same tissue types profiled across different platforms, variations in cell density and gene panel size, and whole-slide inputs. Together, these evaluations provide a systematic understanding of the strengths and limitations of existing methods and establish a foundation for evidence-based method selection.

Our benchmark revealed that methods integrating transcript coordinates generally achieved higher segmentation accuracy but were often limited in throughput and scalability. Methods leveraging existing segmentation priors showed greater accuracy, while those without priors offered broader applicability but often exhibited reduced performance. No single method combined high accuracy with broad applicability, highlighting the importance of selecting methods according to data characteristics and downstream analysis requirements. We also observed that segmentation errors could propagate into downstream analyses, particularly through transcript misassignment and contamination between neighboring cells. We further identified key factors underlying these limitations, including transcript sparsity, stringent quality-filtering thresholds, the computational costs of graph construction and community detection, and transcript contamination at cell boundaries, providing directions for future algorithm development.

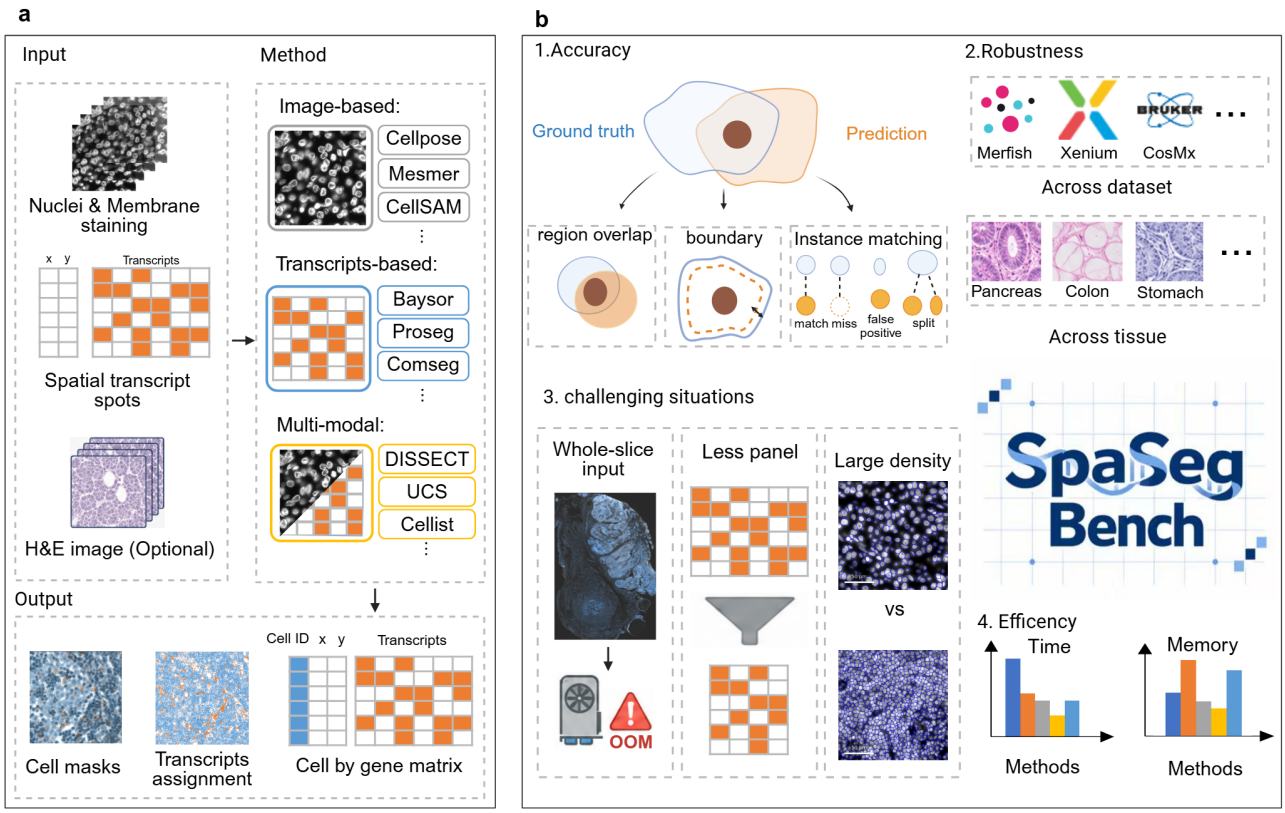
To translate these findings into practical guidance, we developed an interactive web platform for SpaSegBench that integrates two complementary functionalities. First, it provides comprehensive benchmarking results, enabling researchers to explore and compare method performance across datasets, platforms, and evaluation metrics. Its standardized evaluation pipeline also supports the incorporation of new methods and datasets, allowing continuous expansion of the benchmark. Second, it offers an evidence-based recommendation system that suggests suitable segmentation methods based on user-specified data characteristics, available computational resources, and downstream analysis requirements. These recommendations are informed by generalizable patterns identified in the benchmark, including the effects of segmentation priors, staining types, and cell density on accuracy, as well as the relationship between transcript abundance and computational cost. Together, SpaSegBench serves as an extensible resource for evaluating segmentation methods and guiding their selection across diverse ST applications.

2 Results

2.1 Overview of the benchmarking framework

Here we compared 16 cell segmentation methods(Fig. 1a). Depending on what kinds of input they need for segmentation, we first divided them into transcript-based, image-based and multimodal-based methods like Fig. 1a. Then we classified them along two independent axes. By algorithm, they comprise image-based models (Cellpose-SAM[23], Cellpose-3[24], CellSAM[25], CelloType[26], Mesmer[27], StarDist[28], and UNSEG[29]), statistical point-cloud models (Baysor[30], BOMS[31], and ProSeg[32]), graph-based models (Bering[33] and ComSeg[34]), and image-guided multimodal methods (UCS[35], Cellist[36], GeneSegNet[37], and DISSECT[38]). By prior dependence, ProSeg, UCS, and Bering require an existing segmentation as a prior, whereas all other methods do not, although Baysor and ComSeg can optionally use one. To avoid introducing extraneous variables, we converted each dataset into the input format officially required by each method and scored all outputs with the same evaluation pipeline (Supplementary Fig. S3).

We evaluated the methods along six lines (Fig. 1b). Besides segmentation accuracy, we assessed robustness across platforms and tissues, performance in challenging scenarios, computational efficiency, and the impact of segmentation errors on downstream analysis, and we derived practical guidance from these results. Segmentation accuracy was assessed with metrics of boundary agreement, instance matching, transcript assignment, count fidelity, expression similarity, and localization error (Methods). The results of these analyses are provided in Supplementary Tables S2 to S6. We assembled 31 publicly available datasets from published studies and public databases, covering multiple species and tissues (Fig. 1c and Supplementary Table S1). The datasets were generated with MERFISH[3], Stereo-seq[2], STARmap[4], CosMx WTx[39], CosMx 6K[39], Xenium[5] and Xenium Prime 5K[5], and were selected to span a range of computational and biological conditions, including dataset scale, data completeness,



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No.	Platform	Species	# Datasets	Resolution	# Cells	# Genes	# Transcripts
1	Xenium (500-plex)	Human	5	≤1 cell	265,007	541	55,613,344
2	Xenium Prime 5K	Human Mouse	5	≤1 cell	155,419	8,665	134,600,556
3	CosMx 6K	Human	2	≤1 cell	1,020,816	6,398	788,375,832
4	CosMx WTx	Human	1	≤1 cell	48,944	18,946	65,794,237
5	MERFISH (500-plex)	Human Mouse	13	≤1 cell	572,807	499	271,145,872
6	Stereo-seq (WT)	Mouse	4	0.5 μm/bin	302,896	22,588	277,764,869
7	STARmap 1K	Mouse	1	≤1 cell	1,424	1,019	471,295

Figure 1. Overview of the benchmark. a, Overview of the cell segmentation workflow. Image-based, transcript-based, and multimodal methods take nuclear and membrane staining, transcript coordinates, or H&E images as input and return cell masks, transcript-to-cell assignments, or cell-by-gene matrices. **b**, Evaluation framework, covering segmentation accuracy (region overlap, boundary agreement, and instance matching), robustness across platforms and tissues, challenging scenarios (whole-slide inputs, gene panel size, and cell density), and computational efficiency (runtime and memory). **c**, Datasets from the Xenium, CosMx, MERFISH, Stereo-seq, and STARmap platforms, summarized by species, number of datasets, spatial resolution, and numbers of cells, genes, and transcripts.

spatial resolution, species and tissue of origin, number of marker genes, inter-slice similarity of gene expression, and intra-slice heterogeneity[40].

2.2 Overall performance in conventional scenarios

We first summarized the performance of each method as a composite cross-platform rank across all metrics (Fig. 2a and Methods), and unless otherwise stated, the values below are medians across datasets. Among the transcript-based methods, ProSeg and UCS reached the highest detection F1 (0.998 and 0.997) and overall correctness (0.875 and 0.818), ProSeg had the best count fidelity (count Pearson correlation 0.915), ComSeg combined high detection accuracy with low localization error (F1 0.940 and 1.25 μm), while Cellist and BOMS ranked lowest. Among the image-based models, Mesmer, Cellpose-SAM, and Cellpose-3 had the highest composite ranks, and CelloType and UNSEG the lowest. However, no model exceeded a boundary F1 of 0.112, although Dice reached 0.793.

To explain the strengths and weaknesses of each method in depth, we examined the four algorithmic categories in turn and then the role of the prior. Each algorithmic category showed a characteristic profile of strengths and weaknesses (Supplementary Tables S2 and S3). Among the image-based models, segmentation quality was similar across models, whereas detection quality varied widely. For example, StarDist predicted the number of cells most faithfully, CelloType under-segmented, and UNSEG over-segmented with similar segmentation qualities compared to methods in other algorithmic categories. Among the statistical point-cloud models, the transcript assignment rates all remained relatively high, but their accuracy varied widely. For example, ProSeg was the most accurate but the accuracy of BOMS was very low (0.242), and Baysor tended to over-segment. Among the graph-based models, they both had poor count fidelity but ComSeg assigned transcripts unstably across datasets and Bering detected cells less accurately. Among the image-guided multimodal methods, they typically excel in precision, purity, or accuracy, but often performed badly on recall and the number of assigned transcripts. DISSECT produced the purest cells of all 16 methods in the downstream analysis (Supplementary Fig. S2b) but assigned fewer transcripts. Cellist had the highest precision of all methods without prior but a low recall, and completed only 16 of the 31 datasets (Supplementary Fig. S3).

Beyond the algorithmic category, whether a method started from the platform segmentation strongly affected its accuracy. The two most accurate methods, ProSeg and UCS, both use this prior, and they were the only two methods whose count Pearson correlation reached 0.7 on most datasets. Because most reference annotations were also derived from the platform segmentation, we tested all 16 methods on a manually annotated dataset of 2,442 cells to rule out circular evaluation (Supplementary Tables S2 and S3). UCS and ProSeg again reached the highest F1 (0.990 and 0.977), so their advantage is not merely an artifact of circular evaluation. Within the statistical point-cloud category, the prior explained most of the difference between ProSeg and the other two methods, and Baysor itself improved markedly when given a prior on the Xenium pancreatic cancer dataset (F1 0.894 versus a median of 0.732 without one). A prior alone did not guarantee accuracy, however, because Bering uses the same prior as ProSeg yet reached an overall correctness of only 0.222. Prior-based methods are also limited in applicability, because they cannot be used without a platform segmentation, cannot recover cells that the prior misses, and are bounded by the quality of the prior.

To find out which metrics carry independent information and can therefore serve as the foundation of our extensible evaluation framework, we analysed the rank consistency among metrics (Fig. 2b,c and Methods). Within each dataset, we oriented error and distance metrics so that larger values indicate better performance, ranked the methods on every metric (Supplementary Tables S2 and S3), and computed the Spearman rank correlation for every pair of metrics. The rank correlations revealed three patterns. First, precision and recall were only moderately correlated for both the transcript-based methods and the image-based models, and several methods showed contrasting values of the two metrics, such as the high precision and low recall of Cellist, consistent with its systematic under-segmentation. Second, two further dimensions carried largely independent information. For the transcript-based methods, assignment accuracy had the lowest correlations with F1, precision, and recall of all pairs of metrics, and for the image-based models, segmentation quality was only weakly related to the region-overlap metrics. Third, apart from these exceptions, metrics within the same capability family were highly redundant. The localization, expression, and count metrics of the transcript-based methods were strongly correlated within their families, and for the image-based models, Dice and intersection over union (IoU), and instance F1 and detection quality, ranked the models identically. Thus, a correctly detected cell does not necessarily retain its own transcripts or expression profile, and F1 alone cannot reveal whether a method over- or under-segments, so a comprehensive assessment should rely on multiple complementary metrics. In addition, given the practical demands of real applications, runtime and peak memory should also be reported.

By ranking all 16 methods together on the four detection metrics shared by both groups, we got a consistent picture (Fig. 2d). ProSeg, UCS, and ComSeg again ranked highest, and UNSEG, CelloType, Cellist, and StarDist lowest. Cellist, GeneSegNet, and DISSECT had the most uneven profiles, with a difference of 0.50, 0.50, and 0.49 between their best and worst metric, because each combined a relatively high precision with a poor cell count ratio (Fig. 2d). Notably, ProSeg and UCS belong to

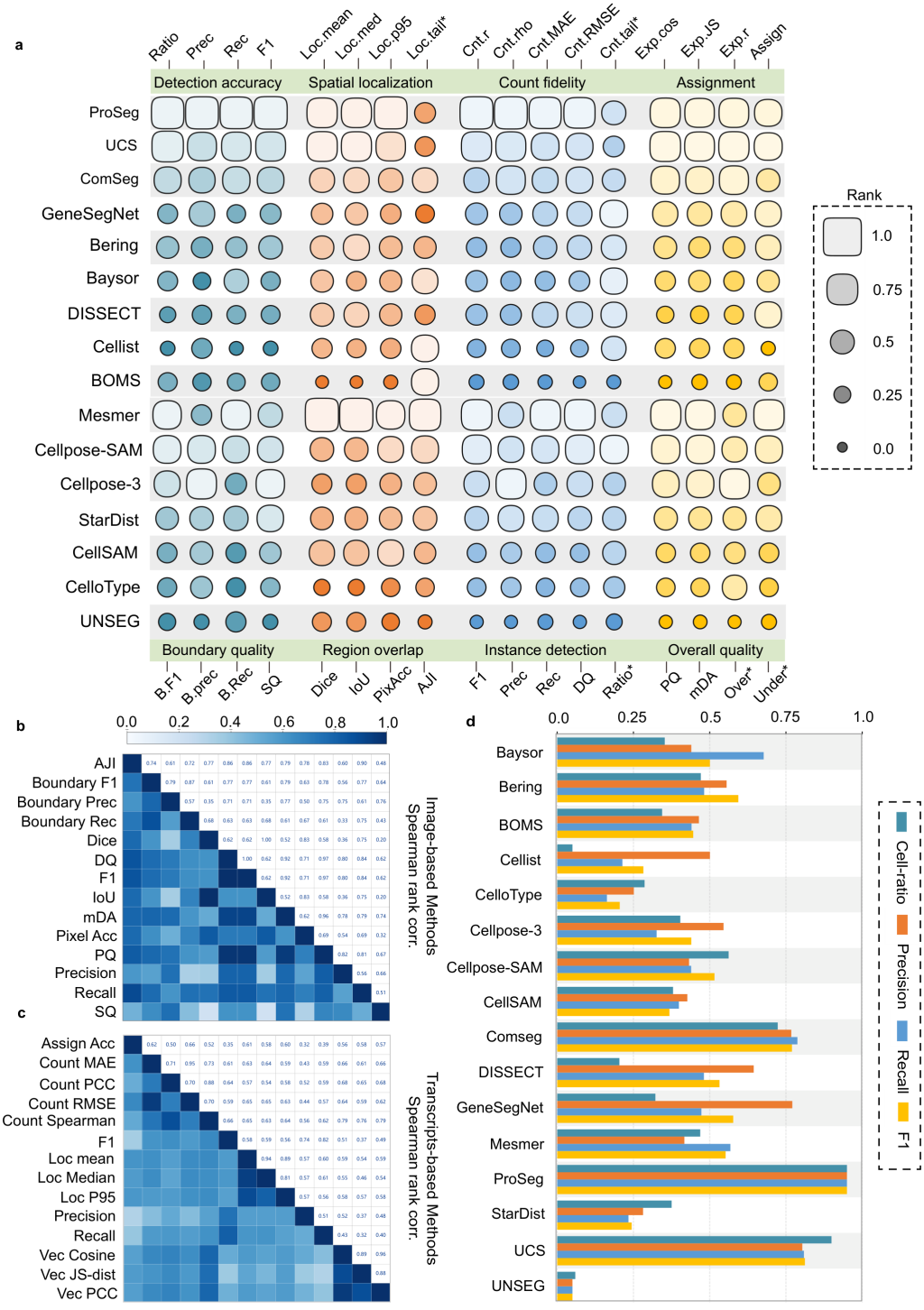

Figure 2. Overall performance of the segmentation methods and consistency among evaluation metrics. a, Composite cross-platform ranks of transcript-based (top) and image-based (bottom) methods on all metrics, computed as described in Methods. Larger and more square markers indicate better ranks, and colours denote metric categories. Asterisks mark derived metrics. **b,** Spearman rank correlations among mask-level metrics of image-based models, computed within each dataset and averaged across datasets. **c,** As in **b**, for transcript-level metrics of transcript-based methods. **d,** Normalized ranks of all 16 methods on the four detection metrics shared by both groups.

different algorithmic categories but both start from the platform segmentation, suggesting that performance depends both on how a method forms cells and on whether it uses a prior.

2.3 A case study confirms the category-specific patterns

To confirm these findings and illustrate them more directly, we further examined on the Xenium breast cancer dataset the characters of strengths and weaknesses of each category described above (Fig. 3a,b and Supplementary Table S4). ComSeg and UCS reproduced the reference transcript-to-cell partition most closely (adjusted Rand index (ARI) 0.728 and 0.682, with assignment rates of 0.953 and 0.974). They were followed by the image-based CellSAM and Cellpose-SAM (0.676 and 0.665), which assigned fewer transcripts (0.752 and 0.650) but rarely mixed transcripts from different cells (homogeneity 0.972 and 0.978). A high assignment rate did not imply an accurate partition. BOMS assigned almost every transcript yet had one of the lowest ARI (0.143) and the lowest homogeneity (0.742), consistent with the merging of neighbouring cells by mean shift, whereas DISSECT and Bering lost most transcripts (assignment rates 0.034 and 0.456), which lowered their completeness (0.613 and 0.705).

The same pattern held across the ten Xenium datasets (Fig. 3c,d). ProSeg, UCS, and ComSeg reached a median F1 of 0.996, 0.990, and 0.989 and the highest count fidelity (count Pearson correlation 0.889-0.900), whereas Bering preserved per-cell counts poorly (0.060). DISSECT combined a high assignment accuracy (0.935) with a low F1 (0.403), because the few cells it detected were assigned correctly. Among the image-based models, Mesmer, Cellpose-3, Cellpose-SAM, and CellSAM had similar F1 (0.668-0.673) and Dice (0.806-0.820), and their median boundary F1 on Xenium (0.183-0.215) was about twice their median across all platforms. Spatial maps of transcript-based clusters showed the same patterns on this and nine further datasets (Supplementary Figs. S4- S13).

2.4 The effects of segmentation on downstream analysis

Neighbour contamination is a major problem in current ST analysis. The quality of cell segmentation directly determines the severity of neighbour contamination and has a substantial impact on downstream analysis indirectly. To trace how segmentation errors reach downstream analysis, we used the Xenium clear cell renal cell carcinoma (ccRCC) dataset, for which co-detection by indexing (CODEX) imaging of the same section provides protein-defined cell identities that are independent of any segmentation (Fig. 4). We focused on T cells and myeloid cells, two lineages whose marker genes are well separated (Fig. 4a).

We analysed the effect of segmentation at two levels (Fig. 4b,c). At the level of cells, the number of evaluable protein-identified cells varied widely between methods, from 395 (DISSECT) to 2,782 (Cellpose-3) for T cells and from 16 (DISSECT) to 1,585 (Baysor) for myeloid cells (Fig. 4c and Supplementary Table S5). At the level of transcripts, contamination was substantial for almost every method. The median ratio of foreign to own lineage-specific transcripts ranged from 0.46 (DISSECT) to 1.08 (Bering) in T cells, from 0.23 (DISSECT) to 0.42 (BOMS and UCS) in myeloid cells, and from 1.60 (Baysor) to 2.25 (Cellist) in tumour and epithelial cells (Supplementary Fig. S2b). This contamination was concentrated in particular regions of the RNA embeddings and can therefore blur lineage identity (Fig. 4c and Supplementary Fig. S5). Resistance to contamination was determined mainly by the extent of the predicted masks, as the masks of ProSeg and Cellpose-SAM illustrate (Fig. 4b). Across all methods, the median foreign/own ratio correlated with the fraction of the region covered by cells (Spearman $\rho = 0.76$ for T cells and 0.93 for myeloid cells). The three methods with the smallest, nucleus-like masks (DISSECT, CelloType, and UNSEG) covered 4-25% of the region and were the most resistant (median ratio 0.46-0.57 in T cells and 0.23-0.30 in myeloid cells), but they recovered fewer evaluable cells (Supplementary Fig. S2b). In contrast, the most expansive methods, UCS and BOMS, covered 85-90% of the region and, together with Bering, were the least resistant (1.00-1.08 in T cells and 0.36-0.42 in myeloid cells).

Beyond the extent of the masks, contamination depended on the local environment of each cell and on the composition of the tissue. Within each cell, the ratio of foreign to own transcripts increased with the density of foreign-lineage transcripts in the surrounding 15 μm (Fig. 4d and Supplementary Table S5), indicating that contamination arises mainly from neighbouring cells. For T cells, the slope of this relationship was 0.89-4.02 across 15 methods, and we did not interpret the slope of BOMS, whose T cells were already heavily contaminated in sparse neighbourhoods. For myeloid cells, the slope was generally smaller (0.39-3.04), with the exception of DISSECT, which retained only 16 myeloid cells. Accordingly, the sources of contamination were consistent across methods and reflected the composition of the tissue (Fig. 4e and Supplementary Fig. S2a,c). For T cells, myeloid cells were the largest source, contributing 30-40% of foreign transcripts, followed by endothelial (19-22%) and plasma-cell transcripts (13-25%). For myeloid cells, endothelial cells were the largest source (25-34%), followed by T-cell (17-34%) and tumour or epithelial transcripts (12-23%). Spatially, contamination was concentrated at and beyond the cell boundary (Fig. 4f). For every method, moving the boundary 3 μm outward increased the number of foreign transcripts per cell more than that of own transcripts, so small errors in boundary placement translate directly into contamination. For example, the T cells of BOMS already contained 41.0 foreign transcripts per cell at the boundary, in line with their large cell sizes.

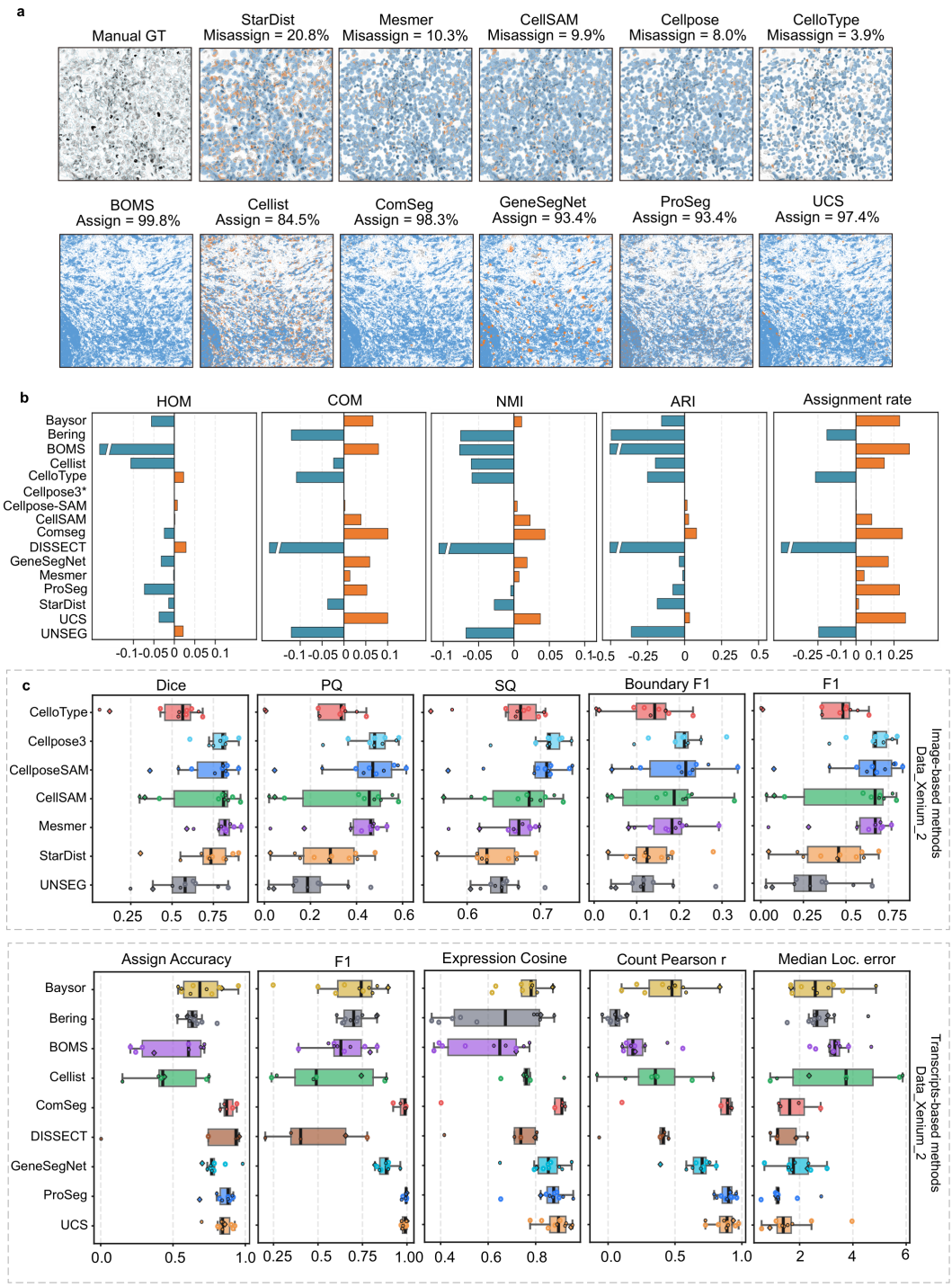

Figure 3. A case study on xenium breast cancer dataset and segmentation performance across the ten Xenium datasets. **a,** Segmentation of each method. For image-based models, light blue and orange mark mask regions that agree and disagree with the reference, and for methods that use transcripts, light blue, orange, and grey mark correctly assigned, incorrectly assigned, and unassigned transcripts. **b,** Agreement between each method's transcript-to-cell partition and the reference in a 1,062.5 × 1,062.5 μm region of the same dataset, measured by HOM, COM, NMI, ARI, and assignment of transcripts. Bars show differences from Cellpose-3 (asterisk). **c,** Dice, panoptic quality (PQ), segmentation quality (SQ), boundary F1, and instance F1 of the seven image-based models on the ten Xenium datasets. **d,** As in **c**, for the nine transcript-based methods, with one representative metric per capability family, namely assignment accuracy, detection F1, cosine similarity of expression profiles, Pearson correlation of per-cell counts, and median localization error.

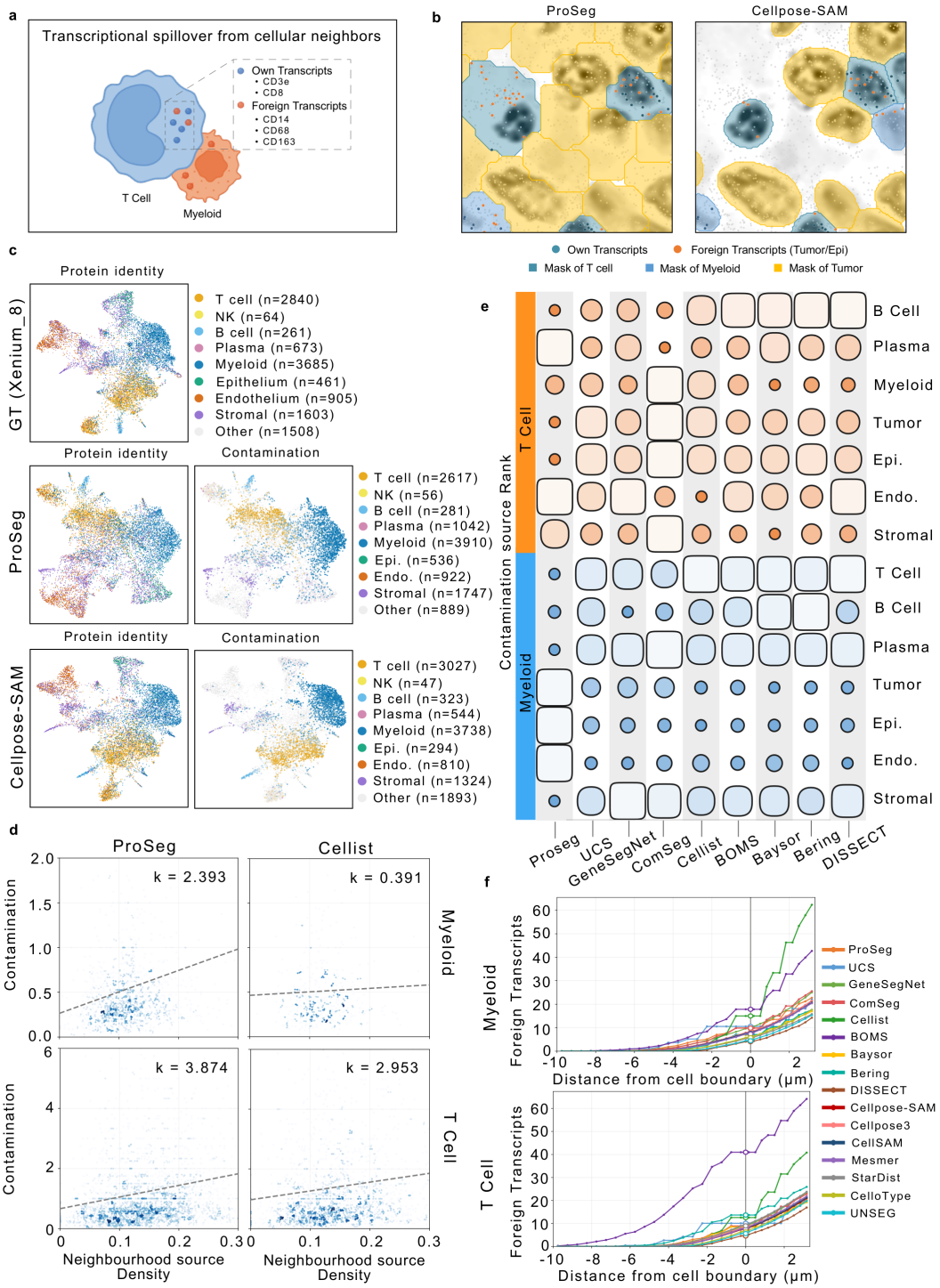

Figure 4. Evaluation on the Xenium ccRCC dataset and impact on downstream analysis. a, Schematic of transcript misassignment between neighbouring cells. Protein-defined lineage identity serves as an independent reference. **b,** The same tissue region segmented by ProSeg and Cellpose-SAM, showing own-lineage and foreign (tumour or epithelial) transcripts and the masks of T, myeloid, and tumour cells. **c,** UMAP embeddings of the cell-by-gene matrices of the GT, ProSeg, and Cellpose-SAM, coloured by protein-defined lineage (left) and by contamination (right), with *n* giving the number of cells per lineage. **d,** Foreign/own transcript ratio of each cell against the density of foreign-lineage transcripts within 15 μm outside it, for ProSeg and Cellist, with dashed linear fits of slope *k*. **e,** Sources of contamination in protein-defined T cells and myeloid cells for the nine transcript-based methods. **f,** Cumulative foreign-lineage transcripts per cell against the signed distance to the cell boundary (negative inside) for all 16 methods, in myeloid cells (top) and T cells (bottom).

Together, these results show that the effect of a segmentation error depends jointly on the segmentation strategy, the extent of the predicted boundary, the transcript abundance of the cell, and its local cellular context. Larger cells retain more of their own transcripts but take up even more from their neighbours, and smaller masks reduce contamination only at the cost of losing cells, so cell recovery, transcript retention, and transcript purity have to be traded off against one another. This also explains why assignment accuracy, rather than assignment rate alone, matters for downstream analysis, because a method that assigns more transcripts helps only if it assigns them to the correct cell. Almost all 16 methods therefore still have room for improvement in addressing neighbour contamination.

2.5 Robustness and performance in challenging scenarios

Segmentation accuracy was largely independent of dataset size, which ranged from 1,365 to 190,965 cells across the analysis regions (Supplementary Table S6a), but it depended strongly on the platform. Apart from ProSeg and UCS, the methods showed distinct platform preferences (Fig. 5a). To separate platform effects from tissue effects, we compared the same tissues profiled on different platforms and different tissues profiled on the same platform (Supplementary Fig. S1c,d and Tables S6b-d), excluding the haematoxylin and eosin (H&E)-based Xenium pancreatic cancer dataset for image-based models. The platform effect exceeded the tissue effect for 10 methods, whereas the tissue effect dominated for BOMS, ComSeg, Cellpose-SAM, and StarDist. Within platforms, prior-based methods were the most stable across tissues, and among methods without an external prior, Baysor was the most consistent across MERFISH tissues and GeneSegNet across Xenium tissues. Across platforms, prior-based refinement generalized best and learned multimodal methods worst, and among image-based models, the generalist Cellpose-SAM transferred better than Mesmer and CellSAM. Consistent with these effects, the rankings of methods agreed more closely between datasets from the same platform than between datasets from different platforms (Supplementary Fig. S1a,b).

We then examined three challenging scenarios. Cell density revealed two opposite failure modes (Fig. 5b and Supplementary Fig. S14). BOMS improved with increasing density, because the density modes on which it depends become unreliable in sparse tissue, whereas GeneSegNet and ComSeg declined, because methods that infer boundaries from transcript proximity or from learned staining patterns tend to merge adjacent cells in dense tissue. These trends persisted within platforms for GeneSegNet and, in MERFISH, for BOMS and DISSECT, whereas prior-based methods were insensitive to density and image-based models did not deteriorate. For gene panel size, the Xenium Prime 5K panels yielded about four times as many transcripts per cell as the standard panels of about 540 genes (median 658 versus 166), yet Baysor and Bering had lower F1 on the larger panels and over-segmented more, whereas ProSeg, UCS, and GeneSegNet were largely unaffected (Fig. 3c,d and Supplementary Table S1). For whole-slide inputs, most transcript-based methods exceeded their own median F1 on the largest dataset, Xenium pancreatic cancer, whereas image-based models performed poorly on it, a decline that reflects its H&E input rather than its size (Fig. 5a). The main challenge of large inputs was instead computational. Most image-based models except UNSEG were fast, the graph-based methods were among the most costly, and Bering had the highest median peak CPU memory, whereas Cellpose-3 had the highest median peak GPU memory (Figs. 5a and 6a). Although most methods could process typical datasets on a standard workstation, peak memory exceeded 250 GiB for Bering, UNSEG, and UCS on at least one dataset (Fig. 5a). For most transcript-based methods, runtime and memory grew mainly with the number of transcripts, whereas for image-based models they grew with image area (Fig. 5b and Supplementary Figs. S15 and S16).

2.6 Summary and Recommandation

Taking all these findings into account, we integrated the results of the benchmark into a step-by-step workflow for cell segmentation (Fig. 6b). In the pre-selection stage, the dataset is processed and loaded, the spatial omics data are preprocessed and quality-controlled, and the completeness of the data is assessed to determine which inputs are available, namely nuclear or membrane staining, H&E images, and transcript coordinates (Fig. 6b, top). In the method-selection stage, the choice of a method follows directly from these inputs (Fig. 6b, middle). When no image is available, transcript-based methods such as ProSeg and ComSeg are used, with ProSeg preferred when a platform segmentation can serve as its prior. When an image is available but transcripts are not, image-based models such as Cellpose-SAM and Mesmer are used, and when both images and transcripts are available, image-guided multimodal methods such as DISSECT and Cellist can combine the two. At this stage, the runtime and peak memory of each candidate method should also be weighed against the available hardware (Fig. 6a). In the post-selection stage, the selected strategy is run and its preliminary result is checked (Fig. 6b, bottom). A result that is not acceptable leads to an adjustment of the inputs or parameters and a new run, whereas an acceptable result is first inspected visually and then evaluated with quantitative metrics. Only when these metrics meet expectations is the segmentation passed on to downstream analysis, and otherwise the strategy is adjusted and run again.

In addition, we defined a compact set of recommended metrics (Fig. 6b, bottom) for the evaluation step of this workflow. Precision and recall should be reported separately, and for transcript-level evaluation, one metric from each of the localization, expression, count, and assignment families is sufficient, for example the median localization error, the cosine similarity of

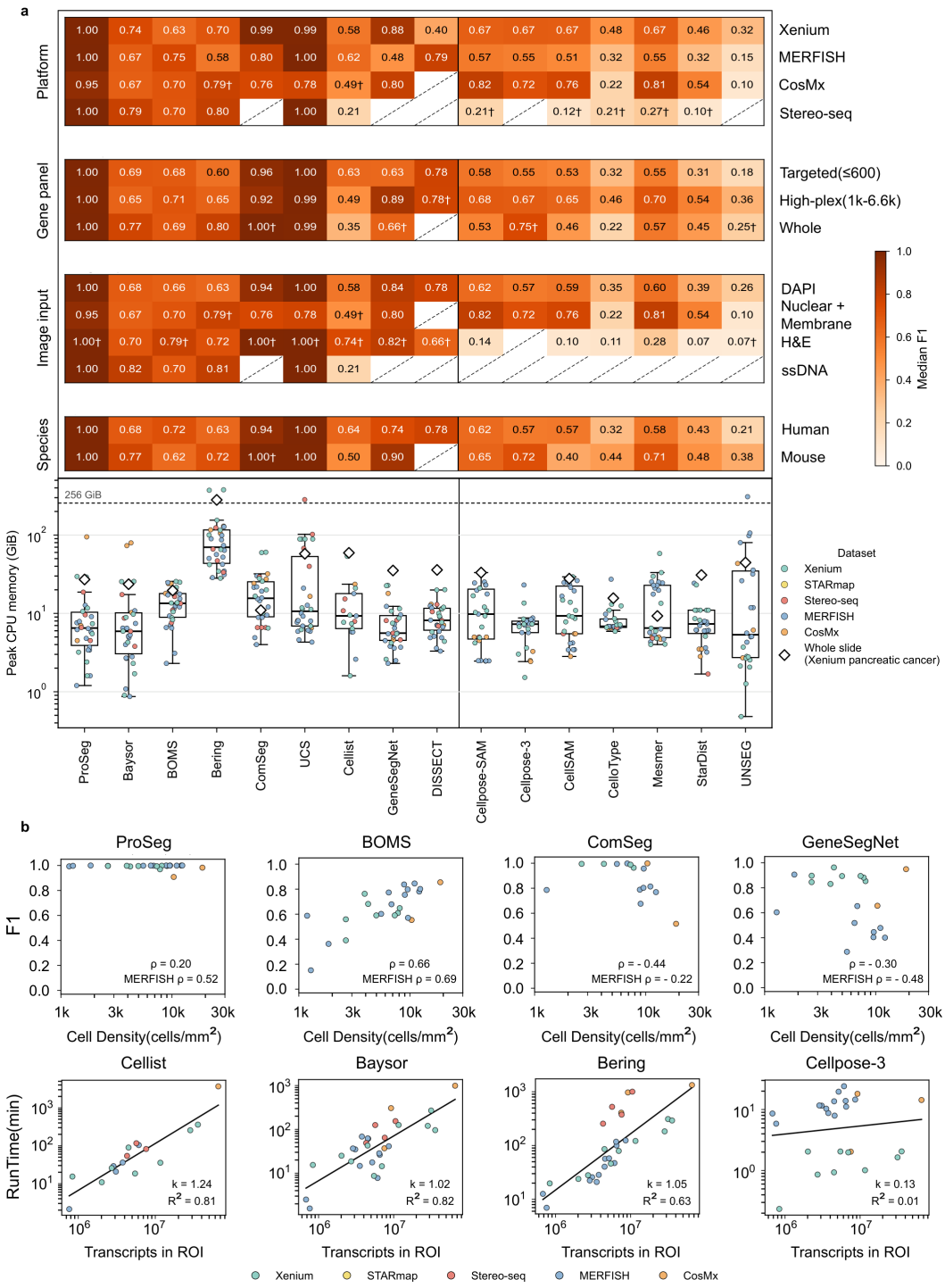

Figure 5. Robustness of segmentation accuracy and computational cost of the 16 methods in challenging and conventional scenarios. a, Top, median F1 of each method within each category of platform, gene panel, image input, and species. Daggers mark single-dataset categories. Bottom, peak CPU memory of each method across its completed datasets on a logarithmic scale, and the dashed line marking the 256 GiB of a typical workstation. Boxes show the median and interquartile range, and whiskers extend to 1.5 times the interquartile range. **b,** Top, F1 against the cell density of the analysis region for ProSeg, BOMS, ComSeg, and GeneSegNet, with the Spearman correlation across all datasets and within MERFISH. Bottom, runtime against the number of transcripts in the analysis region for Cellist, Baysor, Bering, and Cellpose-3, with log–log least-squares fits, scaling exponents k, and R^2. Each point is one dataset.

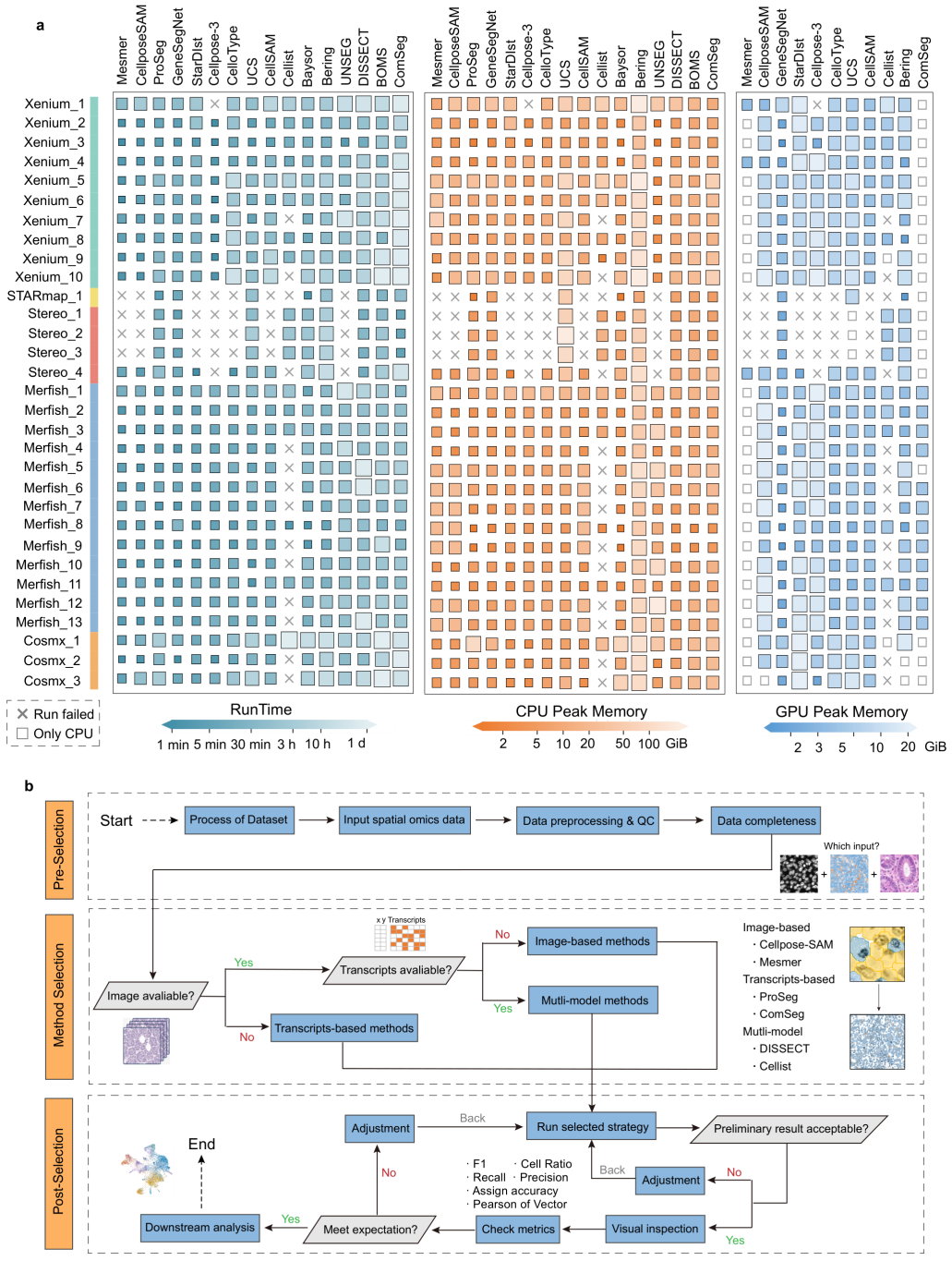

Figure 6. Computational efficiency of cell segmentation methods and a practical segmentation workflow. a, Runtime, peak CPU memory, and peak GPU memory (for methods that use a GPU) of all 16 methods on all 31 datasets. Lighter and larger squares indicate longer runtime and higher peak memory usage of a method on the corresponding dataset. **b,** Step-by-step workflow for cell segmentation, consisting of three main steps: data preparation and preprocessing before segmentation, selection of an appropriate segmentation method, and evaluation of segmentation results and downstream analysis.

expression vectors, the Pearson correlation coefficient (PCC) of per-cell counts, and assignment accuracy. For mask-level evaluation, we recommend panoptic quality together with segmentation quality and one region-overlap metric such as Dice. For all methods, runtime and peak memory should be reported alongside accuracy (Fig. 6a). Finally, we implemented this framework as an extensible web platform. It provides a recommendation tool that implements the workflow in Fig. 6b and suggests suitable methods from the characteristics of the user's data, the available hardware, and the requirements of downstream analysis. Beyond the datasets evaluated in this study, the framework can be applied to new datasets and experimental settings, providing practical guidance for segmentation method selection.

3 Discussion

In this study, we benchmarked 16 cell segmentation methods on 31 datasets for accuracy, robustness, computational cost, and downstream consequences. The methods trace two lines of development. Image-based segmentation has moved from shape-prior models such as StarDist[28] to flow-based models such as Cellpose-3[24] and to models built on large pretrained encoders, such as Cellpose-SAM[23,41], CellSAM[25], and CelloType[26], while UNSEG avoids annotations altogether[29]. This progression improved detection, raising panoptic quality from 0.276 for StarDist to 0.409–0.431 for the generalist models, but not boundary precision, which stayed at a boundary F1 of at most 0.112. The second line assigns molecules to cells, through statistical point-cloud models such as Baysor[30] and BOMS[31], graph-based models such as ComSeg[34] and Bering[33], prior-based refinement in ProSeg[32] and UCS[35], and multimodal fusion in Cellist[36], GeneSegNet[37], and DISSECT[38]. These methods improved transcript-level accuracy most when they refined an existing segmentation, but their cost grew with the number of transcripts and reached hundreds of gigabytes of memory on large inputs. Overall, prior-based refinement was the most accurate but depends on a platform segmentation, image-based models were fast but imprecise at boundaries, transcript-based methods without a prior were less accurate or more costly, and learned multimodal methods transferred poorly across platforms. No method was therefore both highly accurate and broadly applicable.

Every method's segmentation errors always propagate into every downstream analysis. Using protein-defined cell identities, we found that the median T cell of several methods carried as many foreign as own lineage-specific transcripts, and this contamination increased with the density of neighbouring cells and with the extent of the predicted masks. Such contamination can blur lineage identity and be mistaken for new cell states or cellular interactions, particularly in dense immune niches[42]. The choice of a method should therefore follow the downstream goal, with compact masks favoured for analyses that depend on transcript purity, such as studies of cell states or ligand-receptor interactions, and high cell recovery favoured for analyses of cell abundance and tissue composition. For the same reason, assignment accuracy, rather than assignment rate alone, should guide the evaluation of transcript-based methods.

Our results point to concrete directions for improving the algorithms. Statistical point-cloud models depended on fixed assumptions about cell scale and treated every transcript as part of a cell, so adaptive cell-scale priors could reduce the over-segmentation of Baysor, modelling of background transcripts could stop BOMS from assigning transcripts that belong to no cell, and allowing ProSeg to add cells missed by its prior would lessen its dependence on the quality of the prior. Graph-based models detected cells accurately but assigned transcripts unstably and at a high cost, which restricting information exchange across putative cell boundaries and mini-batch or tile-parallel GPU processing could address. Image-guided multimodal methods either filtered cells too aggressively or generalized poorly, so adaptive quality thresholds could recover the cells that Cellist discards, assigning transcripts outside the predicted masks could raise the assignment rate of DISSECT, and domain adaptation or training on several platforms could make learned fusion robust to domain shift. Image-based models captured region overlap but not boundaries, which membrane stains, fine-tuning of generalist models on ST stains, and transcripts as weak supervision could improve, while tiled inference with reduced precision could lower the cost of transformer-based models. For all methods whose cost grows steeply with the number of transcripts, tiling with bounded memory and transcript subsampling are the most direct remedies for large inputs.

Beyond individual algorithms, the field needs broader advances in how segmentation is evaluated and used. First, references derived from platform segmentation make detection scores partly circular, and larger manually curated references and orthogonal references, such as the protein-defined identities used here, would remove this circularity. Second, segmentation should become aware of its downstream use. Contamination measured with lineage-specific transcripts could become a standard quality control[43], and probabilistic models that correct misassigned molecules after segmentation, such as resolVI[42,44], could relax the trade-off between transcript purity and cell recovery by letting expansive masks recover more cells while the correction removes transcripts from neighbouring cells. Per-transcript confidence estimates or soft assignments would further let downstream analyses down-weight uncertain cells[45], and benchmarks should accordingly evaluate biological end points, such as the distinction of cell states, in addition to geometric and assignment accuracy. Third, as datasets reach whole-slide and whole-transcriptome sizes, tiled processing, bounded memory, and GPU acceleration should become standard features, and hybrid

workflows that combine fast image-based models with prior-based refinement could extend the best accuracy to platforms without a native segmentation. Segmentation-free approaches, which infer cell types or tissue structure directly from transcript positions, offer a complementary route that avoids hard boundaries altogether[46,47]. Finally, benchmarks must keep pace with the field, and a continuously updated benchmark with standardized inputs and outputs would allow new methods to be compared fairly as they appear[48,49].

Our study has several limitations. First, the downstream contamination analysis relied on a single dataset with matched protein staining and should be confirmed on more tissues and protein panels. Second, each method was run with its recommended or default parameters, so methods that benefit from dataset-specific tuning or retraining, such as GeneSegNet and DISSECT, may be under-represented. Third, some methods could not be run on every dataset or were evaluated only on a sub-region, and most analyses were restricted to regions of interest (ROI) rather than whole slides. Finally, runtime and memory were measured on shared hardware and may vary with system load.

In summary, cell segmentation is the step at which spatial omics data become single-cell data, so its errors set an upper bound on the reliability of every downstream biological conclusion. Yet the choice of a segmentation method has so far rested largely on convenience, single detection scores, or results reported by developers on their own data. This study replaces that practice with a systematic, multidimensional, and reproducible evaluation. We establish a two-axis taxonomy of 16 methods by algorithmic principle and prior dependence, and we provide a compact set of metrics and a step-by-step workflow that turn our results into practical decisions. We further develop an extensible evaluation and recommendation framework that presents the performance of all methods in separate tracks and recommends suitable methods from user-specified inputs with generally applicable rules. As spatial omics moves towards higher resolution, larger tissues, and more modalities, segmentation will undoubtedly grow in importance, and we expect our benchmark to become an extensible reference for new methods, a practical guide for choosing among them, and a foundation for the next generation of segmentation methods.

4 Methods

4.1 Benchmarked methods

Based on a combination of various factors, such as citation impact, availability of open-source implementation and whether the segmentation task was a central focus of the study, we included 16 cell segmentation methods in the benchmark: Cellpose-3, Cellpose-SAM, CellSAM, CelloType, Mesmer, StarDist, UNSEG, UCS, BOMS, ComSeg, GeneSegNet, ProSeg, Baysor, Bering, Cellist, and DISSECT. To ensure a fair benchmark, we ensured compliance with the following rules in every experiment: No additional preprocessing was applied to the data (e.g., filtering by cell number and gene number, unless required by the default parameters of each method); The parameters of all methods were set according to the official tutorials or documentation of each method. More detailed descriptions are provided in Supplementary Note 3.

4.2 Benchmarked datasets

We assembled 31 ST datasets generated using five platforms: CosMx spatial molecular imaging (3 datasets), MERFISH (13 datasets), Xenium (10 datasets), Stereo-seq (4 datasets), and STARmap (1 dataset). The collection encompasses human healthy and tumour tissues, mouse brain and colon, and developing mouse embryos. Below, we describe the biological characteristics, assay specifications, and spatial regions included in the benchmark. ROI dimensions are reported in pixels in the corresponding source morphology image, or in the registered H&E image for the Stereo-seq whole-embryo sample. Whole-image analysis refers to the complete data supplied for a sample; for a previously extracted region, it refers to the entirety of that region. Multiple fields of view (FOVs) from a CosMx sample are described together as one dataset.

CosMx datasets. Three human formalin-fixed paraffin-embedded (FFPE) datasets were obtained from the public CosMx releases: pancreas, lymph node, and brain. The pancreas dataset was generated using a pre-commercial whole-transcriptome panel targeting 18,946 genes. We retained all available data, comprising 18 FOVs numbered 51–68 and 48,944 cells in the supplied cell metadata. The data include transcript coordinates and identities, cell expression matrices, cell and compartment labels, and composite morphology images.

The lymph node dataset was profiled using the CosMx Human 6K Discovery RNA panel, which targets 6,175 transcripts. We selected FOVs 48 and 261, containing 4,898 and 4,896 vendor-defined cells respectively. The accompanying morphology data include nuclear and membrane-associated staining together with immune and epithelial markers, including CD45, CD68, and pan-cytokeratin.

The brain dataset comprises human hippocampal tissue profiled using a whole-transcriptome panel covering more than 18,000 genes. Its morphology channels include histone, ribosomal RNA, GFAP, and 4',6-diamidino-2-phenylindole (DAPI). We included 29 FOVs: 4, 18, 51, 52, 56, 65, 74, 87, 90, 101, 104, 110, 124, 131, 147, 173, 187, 220, 224, 234, 242, 250, 255, 256,

264, 296, 298, 307, and 339. These FOVs collectively contain 10,359 cells in the supplied metadata. CosMx FOV images have a spatial sampling of approximately 0.12 µm per pixel and typically contain 4,256 × 4,256 pixels.

MERFISH datasets. Thirteen datasets were obtained from Vizgen's public MERSCOPE releases, including 12 human tumor datasets and one mouse brain dataset. The human datasets cover eight cancer types: one breast cancer; two colon cancers; two liver cancers; one lung cancer; one melanoma; two ovarian cancers; two prostate cancers; and one uterine cancer. These FFPE samples were profiled using the 500-gene Human Immuno-Oncology panel. Before ROI extraction, the supplied metadata contained between 254,347 and 993,825 cells per human dataset. Available data include molecule-level transcript coordinates, cell metadata, cell-boundary polygons, DAPI and poly(T) images, and an additional cell-boundary staining channel. We used a central 10,000 × 10,000-pixel ROI from each human tumor image(except the mouse brain dataset, which we used 20000 × 20000-pixel ROI).

The supplied images used for this collection are mosaics of the third z-plane (z3), accompanied by transforms between transcript coordinates in micrometres and mosaic-image coordinates. These transforms indicate image sampling of approximately 0.108 µm per pixel for the human tumors and 0.109 µm per pixel for the mouse brain.

Xenium datasets. Ten datasets were included, comprising nine public 10x Genomics datasets and one dataset from the SPATCH study. The human pancreatic adenocarcinoma dataset was generated using the 377-gene Human Multi-Tissue and Cancer panel supplemented with 97 custom genes, yielding 474 targets. We retained the full supplied pancreatic cancer section, for which the original experiment metadata report 190,965 cells. The human breast cancer dataset corresponds to section S1 and uses a fully custom 280-gene panel. We selected a 5,000 × 5,000-pixel ROI from this image.

The mouse colon dataset was generated from fresh-frozen tissue using the 379-gene Mouse Tissue Atlassing panel; a 6,000 × 6,000-pixel ROI was selected. The non-diseased human liver dataset was profiled using the 377-gene Human Multi-Tissue and Cancer panel. The fresh-frozen mouse brain dataset comprises a coronal hemisphere profiled using a 5,006-gene mouse panel. The prostate dataset (Xenium_7) comprises FFPE human prostate adenocarcinoma, and the skin dataset (Xenium_9) comprises FFPE primary dermal melanoma; both were generated using a development version of the human 5K panel targeting 5,006 genes. The kidney dataset (Xenium_8) comprises FFPE human ccRCC, with 405 RNA targets (377 predesigned and 28 custom targets) and a matched 27-marker protein panel. For the liver, mouse brain, prostate, skin, and kidney datasets, we selected a 10,000 × 10,000-pixel ROI from each image.

The reactive human lymph node dataset was supplied as a pre-extracted region measuring approximately 335 × 335 µm, corresponding to 1,576 × 1,577 image pixels. We extracted this region to make manaul cell boundary annotaions. Its parent experiment used a development 5K panel with 4,624 RNA targets. We used the complete supplied region without further cropping. The remaining dataset (Xenium-10) is a human multi-tissue sample profiled using the Xenium Prime 5K panel. Its supplied expression matrix contains 405,927 cell profiles and 5,001 genes, with accompanying transcript coordinates, morphology images, and an H&E image. A 10,000 × 10,000-pixel ROI was selected for this dataset. Xenium morphology images have a pixel size of 0.2125 µm.

Stereo-seq datasets. Four datasets were included: one whole-embryo dataset from the STOmics demonstration release and three embryonic dorsal-midbrain datasets from the Mouse Organogenesis Spatiotemporal Transcriptomic Atlas (MOSTA). The whole-embryo sample, D05103C4D4, was obtained from an embryonic day 18.5 (E18.5) female C57BL/6 mouse. The original experiment used a 10-µm-thick fresh-frozen section, Stereo-seq T FF v1.3 chemistry, and H&E staining. Available files include spatial expression data, cell-bin assignments, and a registered H&E image. We selected a central 3,000 × 3,000-pixel ROI from the registered image, corresponding to approximately 1.5 × 1.5 mm at 0.5 µm per pixel.

For MOSTA, we selected E14.5_E1S3, E16.5_E2S6, and E16.5_E2S7, representing one section at E14.5 and two sections at E16.5. We used the complete supplied images without additional ROI cropping, together with their corresponding dorsal-midbrain cell-bin gene expression matrices. The three gene expression matrix (GEM) files contain 18,698, 20,654, and 21,370 detected genes and 4,872, 6,229, and 7,960 distinct nonzero cell labels, respectively. Summing the MIDCounts column gives 3,362,914, 5,292,816, and 7,602,540 unique molecular identifier (UMI) counts, respectively. These molecular statistics describe the supplied dorsal-midbrain exports; transcriptomic coverage is therefore defined by those exports even though the complete accompanying image was retained.

STARmap dataset. We included the BY1 mouse primary visual cortex dataset from the 1,020-gene STARmap collection used in the ClusterMap study. The local molecule table contains 471,295 RNA detections, each associated with a gene identity and three-dimensional spatial coordinates. A corresponding DAPI image measuring 6,984 × 13,820 pixels is available. We retained the full supplied image and molecule table without ROI cropping.

4.3 Evaluation metrics

We evaluated segmentation methods from three perspectives: image-based segmentation quality, cell reconstruction quality for transcripts-based and multimodal methods, and the effect of segmentation on downstream analyses. Image-based methods were assessed from instance masks. Transcripts-based and multimodal methods were assessed from matched cells, molecular profiles and transcript assignments. Downstream evaluation quantified the agreement between transcript partitions generated by each method and the reference segmentation. Unless stated otherwise, larger similarity scores indicate better agreement and smaller distance or error values indicate better performance.

4.3.1 Evaluation metrics for image-based methods

For image-based methods, the predicted instance mask was compared directly with the reference mask. Let G and P denote foreground pixels in the reference and prediction, and let G_i and P_j denote individual cell instances.

Foreground and boundary agreement. Foreground overlap was summarized by the Dice coefficient and intersection over union (IoU):

$$\mathrm{Dice} = \frac{2|G \cap P|}{|G| + |P|}, \qquad \mathrm{IoU} = \frac{|G \cap P|}{|G \cup P|}. \tag{1}$$

Pixel accuracy was calculated from foreground and background agreement. Boundary precision, recall and F1 were calculated after applying the same boundary extraction operator to the reference and predicted masks. Boundary pixels were compared at their exact locations, without an additional spatial tolerance.

Instance detection and segmentation quality. Pairwise overlap between reference cell G_i and predicted cell P_j was measured by $\mathrm{IoU}_{ij} = |G_i \cap P_j| / |G_i \cup P_j|$. Candidate pairs with $\mathrm{IoU}_{ij} \geq 0.5$ were assigned one-to-one by Hungarian matching. The number of matched pairs was treated as TP, with unmatched predicted and reference instances counted as FP and FN, respectively. Instance precision, recall and F1 were calculated from these counts using their standard definitions.

Panoptic quality (PQ) combines detection quality (DQ) and segmentation quality (SQ):

$$\mathrm{PQ} = \mathrm{DQ} \times \mathrm{SQ}, \qquad \mathrm{DQ} = \frac{TP}{TP + 0.5FP + 0.5FN}, \qquad \mathrm{SQ} = \frac{1}{TP} \sum_{(i,j) \in \mathscr{M}} \mathrm{IoU}_{ij}, \tag{2}$$

where $\mathscr{M}$ is the set of matched instance pairs. SQ was set to zero when no valid match was found.

The aggregated Jaccard index (AJI) was used to summarize global instance overlap while accounting for unmatched cell area:

$$\mathrm{AJI} = \frac{\sum_{(i,j) \in \mathscr{M}_0} |G_i \cap P_j|}{\sum_{(i,j) \in \mathscr{M}_0} |G_i \cup P_j| + \sum_{i \in \mathscr{U}_G} |G_i| + \sum_{j \in \mathscr{U}_P} |P_j|}, \tag{3}$$

where $\mathscr{M}_0$ denotes matched overlapping instances and $\mathscr{U}_G$ and $\mathscr{U}_P$ denote unmatched reference and predicted instances.

To assess sensitivity to the matching threshold, detection accuracy $TP/(TP+FP+FN)$ was calculated at IoU thresholds from 0.50 to 0.95 in increments of 0.05. The mean across these ten thresholds was reported as $\mathrm{mDA}_{0.50:0.95}$. This score summarizes threshold dependent detection agreement and does not use confidence ranking.

4.3.2 Evaluation metrics for transcripts-based and multimodal methods

Methods that provide cell centroids, cell annotations, molecular counts or transcript assignments were evaluated after establishing one-to-one correspondence between predicted and reference cells.

Cell matching and detection. Predicted and reference cells were matched by centroid distance. For predicted centroid $\mathbf{p}_i$ and reference centroid $\mathbf{g}_j$,

$$d_{ij} = \|\mathbf{p}_i - \mathbf{g}_j\|_2, \qquad r = 1.25\, Q_{0.95}\left(\{d_i^{\mathrm{NN}}\}\right), \tag{4}$$

where r is the default matching radius and d_i^{NN} is the nearest neighbour distance of reference cell i. Only pairs with $d_{ij} \leq r$ were retained as candidates. The sparse candidate graph was decomposed into connected components and minimum distance Hungarian matching was applied within each component. Detection precision, recall and F1 were then computed from the matched, unmatched predicted and unmatched reference cell counts. We also recorded the difference and ratio between predicted and reference cell numbers. Localization error was summarized by the mean, median, 95th percentile and maximum centroid distance of matched pairs.

Cell type and quantitative feature agreement. When cell type labels were available, matched cells were evaluated using accuracy, macro F1, weighted F1 and balanced accuracy. These measures capture overall label agreement while reducing sensitivity to class imbalance.

For quantitative cell features, agreement across matched cells was summarized by Pearson correlation, Spearman correlation, mean absolute error (MAE) and root mean squared error (RMSE). For paired values x_k and y_k,

$$\begin{aligned} r &= \frac{\sum_k (x_k-\bar{x})(y_k-\bar{y})}{\sqrt{\sum_k (x_k-\bar{x})^2}\sqrt{\sum_k (y_k-\bar{y})^2}}, \\ \mathrm{MAE} &= \frac{1}{M}\sum_k |x_k-y_k|, \\ \mathrm{RMSE} &= \sqrt{\frac{1}{M}\sum_k (x_k-y_k)^2}. \end{aligned} \tag{5}$$

Spearman correlation was calculated by applying Pearson correlation to ranked values.

Spatial organization of numeric features was evaluated with Moran's I on an eight nearest neighbour graph:

$$I = \frac{N}{W}\frac{\sum_i \sum_j w_{ij}(z_i-\bar{z})(z_j-\bar{z})}{\sum_i (z_i-\bar{z})^2}, \tag{6}$$

where $w_{ij}=1$ when cell j is among the eight nearest neighbours of cell i, and $W=\sum_i\sum_j w_{ij}$. Moran's I was calculated separately for prediction and reference, and their difference was used to assess preservation of spatial autocorrelation.

Local neighbourhood consistency. Local cell type composition was assessed on an eight nearest neighbour graph constructed from the reference coordinates of matched cells. For each cell, predicted and reference neighbour type frequency vectors were compared using Pearson correlation and Jensen Shannon distance. For probability vectors $\mathbf{p}$ and $\mathbf{q}$, with $\mathbf{m}=(\mathbf{p}+\mathbf{q})/2$,

$$\mathrm{JSD}(\mathbf{p},\mathbf{q}) = \frac{1}{2}\sum_c p_c \log_2 \frac{p_c}{m_c} + \frac{1}{2}\sum_c q_c \log_2 \frac{q_c}{m_c}, \qquad \mathrm{JSDist} = \sqrt{\mathrm{JSD}}. \tag{7}$$

We report the mean neighbourhood Pearson correlation, Jensen Shannon distance and Jensen Shannon divergence across valid matched cells. These neighbourhood metrics were interpreted together with detection recall because only matched cells contribute to the graph.

Transcript assignment and molecular consistency. When transcript level outputs were available, transcript assignment rate was calculated as the fraction of transcript rows assigned to a cell. For each matched cell pair, transcript counts were compared using Pearson correlation, Spearman correlation, MAE and RMSE. Gene count vectors were compared using cosine similarity, Pearson correlation and Jensen Shannon distance, and the corresponding scores were averaged across valid matched pairs.

When transcript identifiers were shared between predicted and reference tables, direct assignment accuracy was calculated as

$$R_{\mathrm{assign}} = \frac{N_{\mathrm{assigned}}}{N_{\mathrm{all}}}, \qquad \mathrm{Acc}_{\mathrm{transcript}} = \frac{1}{|\mathscr{T}_\cap|}\sum_{t\in\mathscr{T}_\cap} \mathbb{I}[m(t)=1], \tag{8}$$

where $\mathscr{T}_\cap$ is the set of shared transcript IDs with valid assignments in both tables and $m(t)=1$ when the predicted and reference assigned cells form a matched cell pair.

4.3.3 Impact of segmentation on downstream task

ARI. The adjusted Rand index (ARI) measures agreement between two partitions after correcting for agreement expected by chance. Each transcript was treated as an observation and its assigned cell ID as the partition label. Let n_{ij} be the number of transcripts shared by reference cell i and predicted cell j, with $a_i=\sum_j n_{ij}$, $b_j=\sum_i n_{ij}$ and $n=\sum_{ij} n_{ij}$. ARI is

$$\mathrm{ARI} = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{n}{2}}. \tag{9}$$

ARI reaches 1 for identical partitions and has an expected value of 0 under random partitioning. Unassigned transcripts were given unique labels so that they did not form an artificial common cluster.

NMI. Normalized mutual information (NMI) measures the information shared by the predicted partition P and reference partition T. Arithmetic mean normalization was used:

$$\mathrm{NMI}(P,T) = \frac{MI(P,T)}{\frac{1}{2}[H(P)+H(T)]}. \tag{10}$$

NMI ranges from 0 to 1, with larger values indicating stronger partition agreement. With this normalization, NMI is numerically equivalent to the V measure and is interpreted together with homogeneity and completeness.

HOM and COM. Homogeneity (HOM) measures whether each predicted cell contains transcripts from a single reference cell. Completeness (COM) measures whether transcripts from each reference cell remain together in a single predicted cell:

$$\mathrm{HOM} = 1 - \frac{H(T \mid P)}{H(T)}, \qquad \mathrm{COM} = 1 - \frac{H(P \mid T)}{H(P)}. \tag{11}$$

Both metrics range from 0 to 1. Low HOM with high COM is consistent with cell merging, whereas high HOM with low COM is consistent with cell fragmentation or incomplete transcript assignment. These scores were interpreted together with transcript assignment rate.

An evaluation unit was a selected ROI, an image, or an individual CosMx FOV, according to the dataset. For analyses spanning multiple FOVs, underlying counts were pooled across FOVs before count-derived metrics were calculated (micro aggregation); per-FOV metric values were not averaged. Computational failures were addressed by retrying when GPU resources became available or, when GPU memory remained limiting, by using CPU execution and adjusting memory-intensive parameters.

4.4 Cell lineage annotation

To assign cell identities for downstream visualization of the ST segmentation results, we used CODEX imaging acquired from the same region to construct an independent protein-based lineage annotation scheme. The marker panel included CD3e, CD20, CD138, CD68, CD163, CD11c, CD16, Granzyme B, pan-cytokeratin (PanCK), CD31, α-smooth muscle actin (α-SMA), and Vimentin, which were used to define eight major cellular lineages: T, natural killer (NK), B, plasma, myeloid, epithelial, endothelial, and stromal cells.

For each protein marker, an intensity threshold was first determined automatically within the fixed $10{,}000 \times 10{,}000$-pixel benchmark ROI using Otsu thresholding. Pixels with intensities above the corresponding threshold were considered initially positive. The resulting binary masks were subsequently subjected to a 3×3-pixel morphological opening operation, followed by removal of connected components smaller than 50 pixels to reduce isolated noise and small imaging artifacts. After this preprocessing, the fraction of positive pixels across the entire ROI was calculated for each marker and defined as the marker-specific ROI positive-area density cut-off.

At the single-cell level, we calculated the fraction of each segmented cell area overlapping the positive mask of each protein marker. For marker m in cell i, relative enrichment was defined as

$$E_{i,m} = \frac{f_{i,m}}{d_m}, \tag{12}$$

where $f_{i,m}$ denotes the marker-positive area fraction within the cell and d_m denotes the corresponding marker-positive area fraction across the full ROI. A cell was considered positive for a given marker when $E_{i,m} > 1$, i.e., when the marker-positive fraction within that cell exceeded the ROI-level positive-area density of the same marker. This normalization allowed marker positivity to be assessed relative to the background abundance of each protein signal within the same ROI, rather than using a common absolute intensity or area threshold across markers.

Lineage identities were then assigned according to predefined marker–lineage relationships: CD3e for T cells, CD20 for B cells, CD138 for plasma cells, CD68/CD163/CD11c for myeloid cells, PanCK for epithelial cells, CD31 for endothelial cells, and α-SMA/Vimentin for stromal cells. For multi-marker myeloid and stromal lineages, a lineage was considered eligible when at least one of its associated markers had $E > 1$, and the maximum enrichment ratio among the corresponding markers was used as the lineage score. NK-cell annotation was defined more conservatively: both CD16 and Granzyme B were required to satisfy $E > 1$, and the smaller of the two enrichment ratios was used as the NK lineage score. When a cell satisfied the criteria for multiple lineages, it was assigned to the lineage with the highest lineage score. Cells that did not satisfy the criteria for any lineage were labeled as "other".

For the ROI analyzed in this study, the marker-specific ROI positive-area density cut-offs were 3.2242% for CD3e, 0.1047% for CD20, 55.7164% for CD138, 1.7533% for CD68, 2.9980% for CD163, 2.9020% for CD11c, 3.9624% for CD16, 32.3564% for Granzyme B, 1.2605% for PanCK, 0.3745% for CD31, 0.8772% for α-SMA, and 3.5260% for Vimentin. These thresholds were derived automatically from the same ROI-level protein images and were kept fixed across downstream analyses to ensure a consistent annotation criterion across different segmentation outputs.

In addition, for the analysis of transcript contamination, a stricter, single-marker criterion was used to obtain unambiguous cells: a cell was counted as a T cell, a myeloid cell, or a tumour or epithelial cell when at least 15% of its mask area lay within CD3e-, CD68-, or PanCK-positive regions, respectively, using the same positive regions as above, and cells meeting two criteria were assigned to the marker with the larger positive fraction. Cells with fewer than three own-lineage or ten lineage-specific transcripts were excluded from this analysis.

4.5 Computational environment

Analyses were performed on a server with two Intel Xeon Gold 6458Q processors (64 physical cores), approximately 1 TiB of RAM, eight NVIDIA GeForce RTX 4090 D GPUs (24,564 MiB reported memory per GPU), and Ubuntu 22.04.5 LTS. Methods used their respective software environments and available CPU or GPU resources.

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Author contributions statement

P.G.: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft and Writing – review & editing. Y.Q.: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft and Writing – review & editing. Y.Y.: Conceptualization, Resources, Supervision, Investigation, Methodology and Writing – review & editing. Y.L.: Conceptualization, Resources, Supervision, Investigation, Methodology and Writing – review & editing. K.Z.: Conceptualization, Resources, Supervision, Investigation, Methodology and Writing – review & editing. Y.C.: Conceptualization, Resources, Supervision, Investigation, Methodology and Writing – review & editing. E.C.: Conceptualization, Resources, Supervision, Investigation, Methodology and Writing – review & editing. L.S.: Conceptualization, Resources, Supervision, Investigation, Methodology and Writing – review & editing.

Data availability

Xenium_1 were downloaded from (https://www.10xgenomics.com/datasets/pancreatic-cancer-with-xenium-human-multi-tissue-and-cancer-panel-1-standard), Xenium_2 were downloaded from (https://www.10xgenomics.com/datasets/xenium-ffpe-human-breast-biomarkers), Xenium_3 were downloaded from (https://www.10xgenomics.com/datasets/preview-data-xenium-prime-gene-expression), Xenium_4 were downloaded from (https://www.10xgenomics.com/datasets/xenium-prime-fresh-frozen-mouse-brain), Xenium_5 were downloaded from (https://www.10xgenomics.com/datasets/human-liver-data-xenium-human-multi-tissue-and-cancer-panel-1-standard), Xenium_6 were downloaded from (https://www.10xgenomics.com/datasets/fresh-frozen-mouse-colon-with-xenium-multimodal-cell-segmentation-1-standard), Xenium_7 were downloaded from (https://www.10xgenomics.com/datasets/xenium-prime-ffpe-human-prostate), Xenium_8 were downloaded from (https://www.10xgenomics.com/datasets/xenium-prime-ffpe-human-cervical-cancer), Xenium_9 were downloaded from (https://www.10xgenomics.com/datasets/xenium-prime-ffpe-human-skin), Xenium_10 were downloaded from (https://www.10xgenomics.com/datasets/xenium-protein-ffpe-human-renal-ccrcc), Cosmx_1 were downloaded from (https://brukerspatialbiology.com/resources/cosmx-ffpe-pancreas-dataset-flat-files/), Cosmx_2 were downloaded from (https://brukerspatialbiology.com/resources/cosmx-human-lymph-node-ffpe-dataset-flat-files/), Cosmx_3 were downloaded from (https://objects.liquidweb.services/smi-public/wtx_manuscript/brain_hippocampus/index_hippocampus.html), Stereo_1 were downloaded from (https://db.cngb.org/stomics/datasets/STDS0000058/data), Stereo_2 were downloaded from (https://db.cngb.org/stomics/datasets/STDS0000058/data), Stereo_3 were downloaded from (https://db.cngb.org/stomics/datasets/STDS0000058/data), Stereo_4 were downloaded from (https://en.stomics.tech/col1370/index.html), Merfish_1 were downloaded from (https://info.vizgen.com/mouse-brain-data), Merfish_2 were downloaded from (https://info.vizge

n.com/ffpe-showcase), Merfish_3 were downloaded from (https://info.vizgen.com/ffpe-showcase), Merfish_4 were downloaded from (https://info.vizgen.com/ffpe-showcase), Merfish_5 were downloaded from (https://info.vizgen.com/ffpe-showcase), Merfish_6 were downloaded from (https://info.vizgen.com/ffpe-showcase), Merfish_7 were downloaded from (https://info.vizgen.com/ffpe-showcase), Merfish_8 were downloaded from (https://info.vizgen.com/mouse-brain-data), Merfish_9 were downloaded from (https://info.vizgen.com/mouse-brain-data), Merfish_10 were downloaded from (https://info.vizgen.com/mouse-brain-data), Merfish_11 were downloaded from (https://info.vizgen.com/mouse-brain-data), Merfish_12 were downloaded from (https://info.vizgen.com/mouse-brain-data), Merfish_13 were downloaded from (https://info.vizgen.com/mouse-brain-data), STARmap_1 were downloaded from (https://github.com/wanglab-broad/ClusterMap/tree/main/datasets/STARmap_V1_1020)

Code availability

The code and tutorials are available via GitHub at (https://github.com/Hussar1025/SpaSeg), The interactive web platform for the cell segmentation methods we developed is available at https://wood-321.github.io/SpaSegBench.github.io/.

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Supplementary Information

Supplementary Notes

Supplementary Note1: Comparisons with prior study

Unlike previous cell segmentation studies that typically focused on a single method with limited evaluation scope, our work provides the first unified and systematic benchmark of 16 methods across 31 real datasets and simulated data spanning five platforms (Xenium, MERFISH, CosMx, STARmap, and Stereo-seq). Based on their underlying algorithmic logic, we grouped these methods into four categories. Statistical point-cloud methods (BOMS and Baysor) partition transcripts through density estimation or probabilistic mixture modeling. Graph-based methods (Bering and ComSeg) encode transcript colocalization as graphs and recover cells through node classification or community detection. Prior-dependent and multimodal methods (Cellist, UCS, ProSeg, GeneSegNet, and DISSECT) anchor on platform-provided nuclear segmentation or image priors, and refine cell boundaries by integrating image and RNA information, probabilistic simulation, or learned foreground prediction. Finally, image-based methods include supervised models (Cellpose-SAM, StarDist, Mesmer, CellSAM, CelloType, UNSEG, and Cellpose3), which rely solely on staining images and perform segmentation through instance segmentation networks, star-convex shape priors, foundation models, or image restoration strategies. Previous method papers usually reported only their own superiority on a few datasets and lacked cross-method, cross-platform systematic comparisons. In contrast, through joint evaluation across four semi-independent dimensions—detection, localization, transcript assignment, and counting—we reveal that no single method is optimal across all dimensions, and that performance strongly depends on the morphological information provided by the platform. We further elucidate the characteristic failure modes of each family through mechanistic analyses, such as the tendency of statistical point-cloud methods to undersegment in dense regions, the molecule leakage across cell boundaries in graph-based methods caused by oversmoothing, the constraints of prior-dependent methods imposed by nuclear thresholds or training-set scales, and the extremely low transcript F1 scores of image-based methods under weak staining. These insights have not been systematically revealed by previous single-method papers.

Moreover, we provide the first systematic analysis of downstream effects, showing that segmentation errors contaminate transcriptomes in a lineage- and neighborhood-dependent manner, and revealing the intrinsic trade-off among cell recovery, boundary extent, transcript retention, and transcript purity, whereas previous work has mostly focused only on geometric metrics of segmentation itself. We further designed three practical challenging scenarios (downsampling, using only highly variable genes, and ultra-large-scale datasets), exposing significant limitations of existing methods in accuracy and memory consumption, and proposed future directions such as "segment-then-correct", matched scRNA-seq references, and spatial context awareness. Finally, we provide a systematic evaluation framework, an executable user guide, and an interactive online platform to support method selection based on data characteristics—practical tools and comprehensive guidance that no previous single-method paper has offered. Therefore, relative to previous work, our contributions offer significant advantages in evaluation comprehensiveness, classification completeness, depth of downstream impact analysis, and practical guidance value.

Supplementary Note2: Foundational principles and implementation details of the evaluated methods in our benchmarking framework

CelloType. CelloType is an end-to-end Transformer-based deep learning framework that unifies object detection, instance segmentation, and cell type classification in tissue images as a multitask learning problem, rather than following the conventional two-stage pipeline of segmentation followed by classification. It consists of a Swin Transformer feature extraction module, a DINO module for object detection/classification, and a MaskDINO segmentation module. Through a shared encoder, deformable attention, contrastive denoising training, and a joint loss, it enables segmentation and classification to mutually reinforce each other and outputs segmentation masks, class labels, and confidence scores.

CellSAM. CellSAM is a general-purpose foundation model for cell segmentation based on the Segment Anything Model (SAM). It first jointly trains ViT and CellFinder to automatically detect cells and generate bounding box prompts, and then freezes the ViT and fine-tunes SAM's neck to generate instance segmentation masks. Consequently, it exhibits strong zero-shot and few-shot capabilities and can adapt to the distribution shift from natural images to cellular images.

Cellpose3. Cellpose3 is a one-click image restoration method for noise, blurring, and undersampling degradations commonly encountered in microscopy, designed to improve cell and nuclear segmentation performance. Unlike conventional pixel-level reconstruction, it chains a trainable restoration network with a frozen pretrained Cellpose segmentation network and jointly trains the restoration network using segmentation, perceptual, and reconstruction losses, so that the output images are both favorable for downstream segmentation and perceptually similar to clean images.

Cellpose-SAM. Cellpose-SAM is a foundation model that combines the ViT-L image encoder of the Segment Anything Model (SAM) with the Cellpose flow-field segmentation framework. It removes SAM's prompt encoder and mask decoder and directly

uses the ViT encoder to predict cell probability and horizontal and vertical flow fields, generating instance masks through gradient tracking. This enhances robustness to channel order, size, noise, blur, downsampling, and anisotropic degradation, as well as cross-domain generalization.

StarDist 2D. StarDist 2D is a method for instance segmentation of cells/nuclei based on a star-convex polygon representation, whose core assumption is that each object is star-convex with respect to its center. It employs convolutional neural networks such as U-Net to predict, for each pixel, the object probability and the radial distances to the object boundary along predefined rays. The predicted radial distances are then reconstructed into instance masks through non-maximum suppression and candidate polygon filtering.

Mesmer. Mesmer is a deep learning-based cell segmentation algorithm for tissue images. It uses a two-channel input consisting of a nuclear marker and a membrane/cytoplasmic marker, and employs a ResNet50 feature pyramid network with four prediction heads to separately predict the centroids and boundaries of nuclei and whole cells. Watershed post-processing is then applied to generate instance segmentation masks.

UCS. UCS is a unified deep learning approach for cell segmentation in subcellular spatial transcriptomics (SST) data. It accurately delineates cell boundaries by integrating nuclear staining/nuclear segmentation information with the spatial distribution of transcripts. The method comprises two convolutional neural networks: Foreground Predict Net predicts candidate cellular foreground regions based on the transcript map, while Cell Predict Net outputs the cell assignment for each pixel under the guidance of foreground features and a nuclear soft mask, thereby achieving accurate cell segmentation and improving transcript assignment.

BOMS. BOMS is a mean shift-based cell segmentation method for imaging-based spatial transcriptomics that uses only the spatial locations and gene labels of mRNA molecules and does not rely on auxiliary staining images such as DAPI/Poly(A). It first constructs a Neighborhood Gene Expression (NGE) vector for each transcript based on its k nearest neighbors, and then performs mean shift iterations in a joint domain composed of the spatial domain and the expression domain, driving molecules to converge to density modes and assigning molecules that converge to the same spatial mode to the same cell.

ComSeg. ComSeg is a graph-based segmentation method that operates directly on spatial RNA point clouds and nucleus positions. It constructs a KNN graph in which RNAs are nodes and gene co-expression values are edge weights, employs a modified Louvain community detection approach aided by nuclear landmarks to partition RNAs into local transcriptional communities, then applies Leiden clustering to generate a tissue-scale transcriptomic domain map, and finally assigns RNAs to the nearest cell centroids based on graph geodesic distances.

GeneSegNet. GeneSegNet is a deep learning-based cell segmentation framework for RNA in situ hybridization data. It converts RNA spatial locations into continuous probabilistic localization maps and feeds them together with imaging information into a unified U-Net, thereby jointly leveraging gene expression and image information to infer cell boundaries. GeneSegNet outputs a cell confidence map, a center map, and an offset map from each pixel to its corresponding cell center, based on which it recovers precise instance masks for individual cells. It also adopts a recursive training strategy in which predictions from the previous round are refined by level-set boundary refinement and then used as training labels for the next round, thereby mitigating the impact of noisy annotations on training.

Baysor. Baysor is a cell segmentation method for imaging-based spatial transcriptomics. It models each cell as a distribution composed of the spatial positions and gene identities of molecules, and employs a Bayesian mixture model to jointly optimize transcriptional composition and cell morphology. It can perform segmentation based solely on detected RNA molecules, or incorporate auxiliary segmentation results such as DAPI, poly(A), or membrane staining as probabilistic priors. Furthermore, it utilizes a Markov random field to ensure spatial connectivity and to distinguish background from intracellular molecules.

ProSeg. Proseg is an unsupervised probabilistic cell segmentation method based on the Cellular Potts model. It allows resampling of transcript positions to account for RNA leakage, replaces the contrived energy function in traditional CPMs with a probabilistic model of the observed spatial distribution of transcripts, and optimizes cell boundaries through Metropolis–Hastings sampling, thereby inferring morphologically plausible cell boundaries without requiring training.

Cellist. Cellist is a multimodal cell segmentation method for high-resolution spatial transcriptomics. It employs a scalable sliding-window parallelization strategy, integrates staining images with spatial expression information, first obtains initial nuclear regions using Watershed or Cellpose, and then uses multimodal registration and probabilistic assignment to assign extranuclear transcripts to neighboring cells or the background, with an emphasis on maintaining intracellular transcriptomic consistency rather than merely pursuing morphological boundaries.

Bering. Bering is a graph deep learning-based method for joint cell segmentation and annotation in spatial transcriptomics. It

leverages spatial co-localization relationships among transcripts to simultaneously perform noise-aware cell segmentation and molecular-level cell type annotation in 2D and 3D data. It treats each transcript as a graph node and uses Neighborhood Gene Components (NGCs) as node features, learning node representations via a GCN. For edges, it integrates node embeddings, RBF distance kernels, and optional staining image embeddings to predict intracellular versus intercellular edges. It then applies Leiden community detection to obtain cell boundaries and uses a node classifier to annotate cell types and background noise.

DISSECT. DISSECT is a multimodal cell segmentation method that integrates cytological images with spatial transcriptomic expression profiles for reconstructing single-cell spatial transcriptomes on imaging-based and sequencing-based high-resolution spatial transcriptomics platforms. It employs a pretrained deep generative model to denoise multiscale image features, predicts cell bounding boxes through an instance-aware detection module, and refines segmentation masks within each bounding box by combining image- and transcriptome-derived gradient fields, with adaptive weighting based on Jensen-Shannon divergence.

UNSEG. UNSEG is an unsupervised, training-free method for cell and nucleus segmentation based on a Bayesian-like framework. It leverages two-channel markers for nuclei and cell membranes and distinguishes nuclei, cell membranes, and background through prior spatial probabilities, a visual contrast likelihood, and a posteriori global and local semantic masks.

Supplementary Note3: Detailed benchmark design from input to output

Benchmark design. We included 16 cell segmentation methods in the benchmark: Cellpose-3, Cellpose-SAM, CellSAM, CelloType, Mesmer, StarDist, UNSEG, UCS, BOMS, ComSeg, GeneSegNet, Proseg, Baysor, Bering, Cellist, and DISSECT. These methods represent three major categories of segmentation strategies, including image-based, transcript-based, and multimodal approaches. For each dataset, analyses were performed on a predefined tissue region or field of view (FOV). Morphological images, spatial transcript coordinates, and initial labels were provided as input when required by the corresponding method. The resulting cell segmentations or transcript-to-cell assignments were then evaluated against the reference annotations defined for the same analysis region.

Datasets and reference annotations. To evaluate segmentation at the level of individual cells and transcripts, we included only platforms that resolve transcript positions at subcellular resolution. Spot-based platforms such as Visium and Slide-seq were not considered, because each of their capture spots can contain transcripts from several neighbouring cells, which precludes a fine-grained assessment of cell boundaries and transcript assignment. For each dataset, analyses were conducted within a dataset-specific square region of interest (ROI) to keep memory requirements manageable. The Xenium datasets included morphology images together with transcript, cell, and boundary information, and the cell boundaries and transcript-to-cell assignments provided by the Xenium platform were used as reference annotations. For the xenium_roi_crop dataset, an additional set of manually delineated cell boundaries was also available. For CosMx, analyses were performed on FOV-specific CellComposite images, with the corresponding platform-generated CellLabels masks used as reference annotations. MERFISH datasets combined mosaic morphology images with detected-transcript tables. When pixel-level reference masks were unavailable, which was the case only for the mouse brain dataset, the platform-provided cell-boundary polygons were converted from physical coordinates to the corresponding mosaic pixel coordinates. Stereo-seq datasets were processed from GEM or GEF molecule records together with CellBin-derived cell labels, and the STARmap dataset consisted of three-dimensional spot coordinates and a DAPI image. The dataset identifiers, input modalities, and analysis regions are summarized in Supplementary Table S1.

Method inputs and preprocessing. Preprocessing preserved the coordinate system of each dataset and supplied the modalities required by each method. For Xenium, morphology channels were selected from OME-TIFF images, while transcript coordinates and feature names were mapped to method-specific input fields; quality and control-transcript filters were applied in workflows that specified them. For CosMx, CellComposite images were paired with CellLabels masks by FOV, and transcript-based analyses used FOV identifiers and global-pixel coordinates. For MERFISH, DAPI and available polyT mosaics were cropped to the same ROI; cell-boundary coordinates were transformed from micrometers to mosaic pixels when rasterized masks were needed. For Stereo-seq, GEM-based analyses used the supplied molecule coordinates, whereas image-based GEF analyses verified image–GEF registration and mapped CellBin labels to the selected ROI. For STARmap, spot coordinates, including the z coordinate, were retained for point-based analyses but not used for segmentations.

The methods who needs images used model-specific preprocessing. For example, StarDist normalized DAPI intensities to the 1st and 99.8th percentiles and expanded the resulting nuclear labels by 15 pixels for whole-cell evaluation in our benchmark. CelloType processed an 8-bit, three-channel image in 512-pixel slices with 5% overlap. GeneSegNet combined morphology tiles with spatial gene-expression heat maps. These transformations were applied within their respective method workflows.

Methods requiring an initial segmentation were designated prior-assisted. UCS combined a rasterized gene map with initial labels derived, according to the dataset and configuration, from platform nuclear or cell boundaries, platform cell centroids,

CellBin assignments, or DAPI-derived nuclei. Proseg and Bering used existing cell assignments as initialization in the relevant transcript-based analyses. The prior source was considered when interpreting comparisons with platform-derived references, particularly where the prior and reference originated from the same segmentation. BOMS used dataset-dependent spatial-scale settings (h_s) and, for large transcript tables, spatial patching to satisfy memory constraints.

Segmentation and output processing. Segmentation was performed through each method's inference interface. Image-based methods generally produced integer-labeled instance masks. StarDist first produced nuclear labels, which were expanded as whole-cell masks for the purpose of a fair comparison. Transcript-based methods produced cell polygons, cell-level tables, or transcript-to-cell assignments: Proseg exported polygons and assignment tables; Bering exported transcript assignments; ComSeg exported AnnData and cell-observation tables; and BOMS exported cell locations and assignments in NPZ format. UCS and GeneSegNet exported labeled TIFF masks, while DISSECT exported an integer mask and proposal table. Method-specific evaluators compared predicted masks, cell locations, or transcript assignments with the corresponding reference data for each ROI or FOV based on a unified set of evaluation metrics.

Supplementary Figures

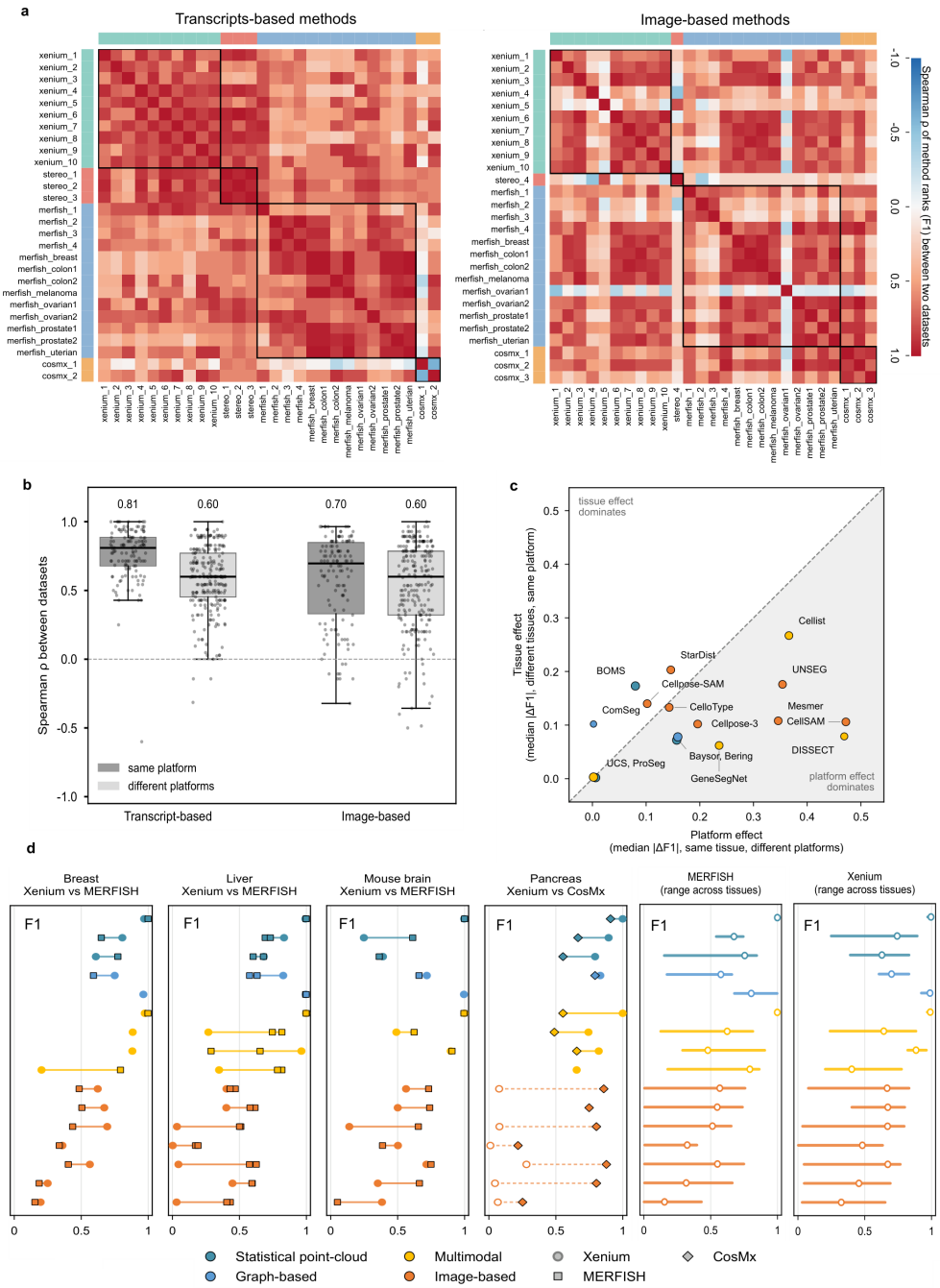

Figure S1. Robustness of method rankings across datasets, platforms, and tissues. a, Spearman correlation between the F1 rankings of methods on every pair of datasets, for transcript-based and image-based methods. Black boxes enclose same-platform pairs. **b,** Correlations from **a** for same-platform and cross-platform pairs. Each point is one pair, numbers above the boxes are medians, boxes show the interquartile range, and whiskers extend to 1.5 times the interquartile range. **c,** Platform effect against tissue effect for each method, with the shaded area below the diagonal marking methods dominated by the platform effect. **d,** Left, F1 of each method on the same tissue profiled on two platforms. Right, range of F1 across tissues within MERFISH (13 datasets) and Xenium (10 datasets).

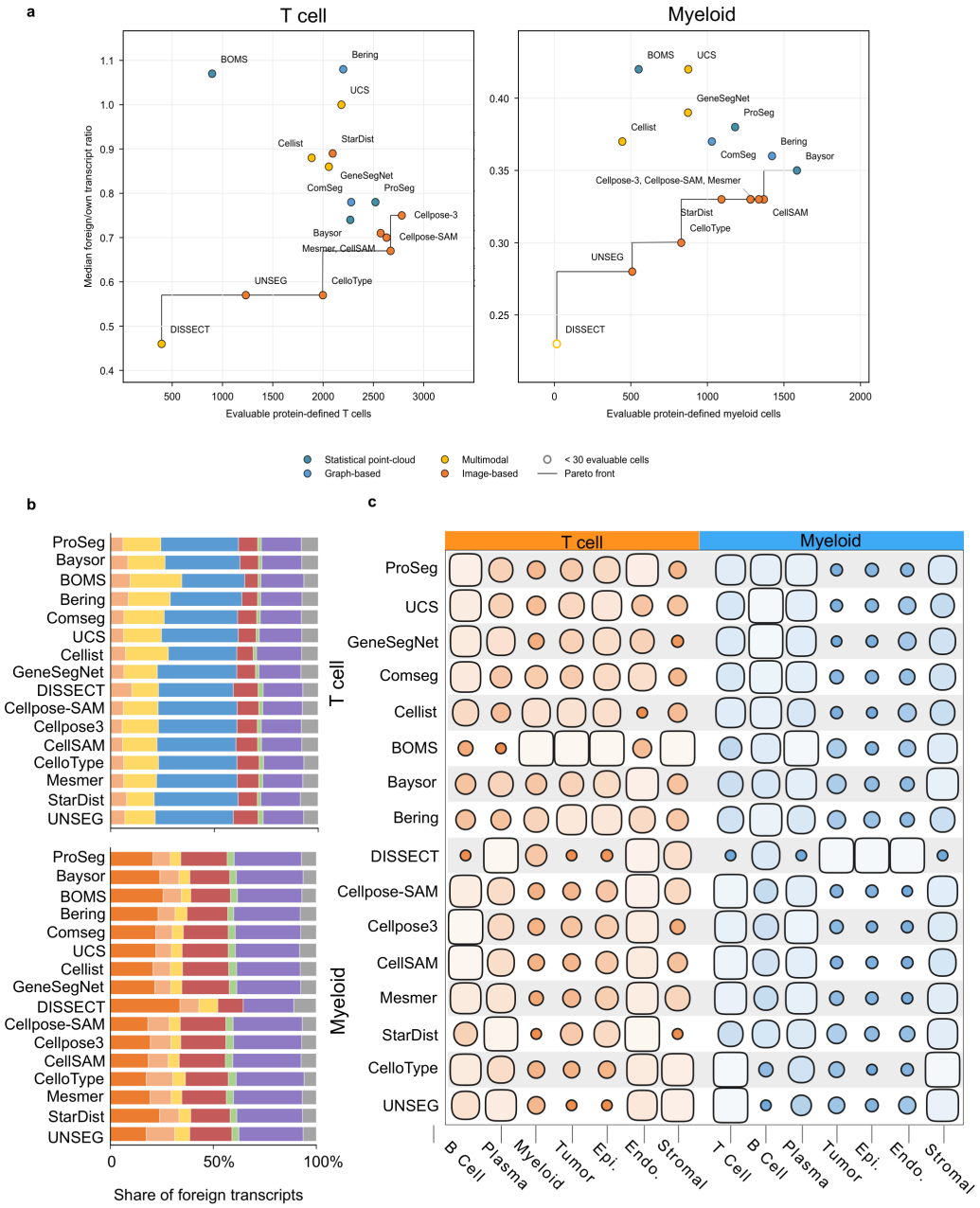

Figure S2. Lineage contamination of protein-defined cells for all 16 methods on the Xenium ccRCC dataset. a, Sources of contamination in protein-defined T cells and myeloid cells, extending Fig. 4e to all 16 methods. Each marker shows the share of foreign transcripts from one source lineage. **b,** Trade-off between cell recovery and contamination. Each point is one method. The open circle marks fewer than 30 evaluable cells, and the black line marks the Pareto front. **c,** Share of foreign transcripts from each source lineage in T cells and myeloid cells for each method. Colours in **b** and of method names in **c** denote algorithmic categories.

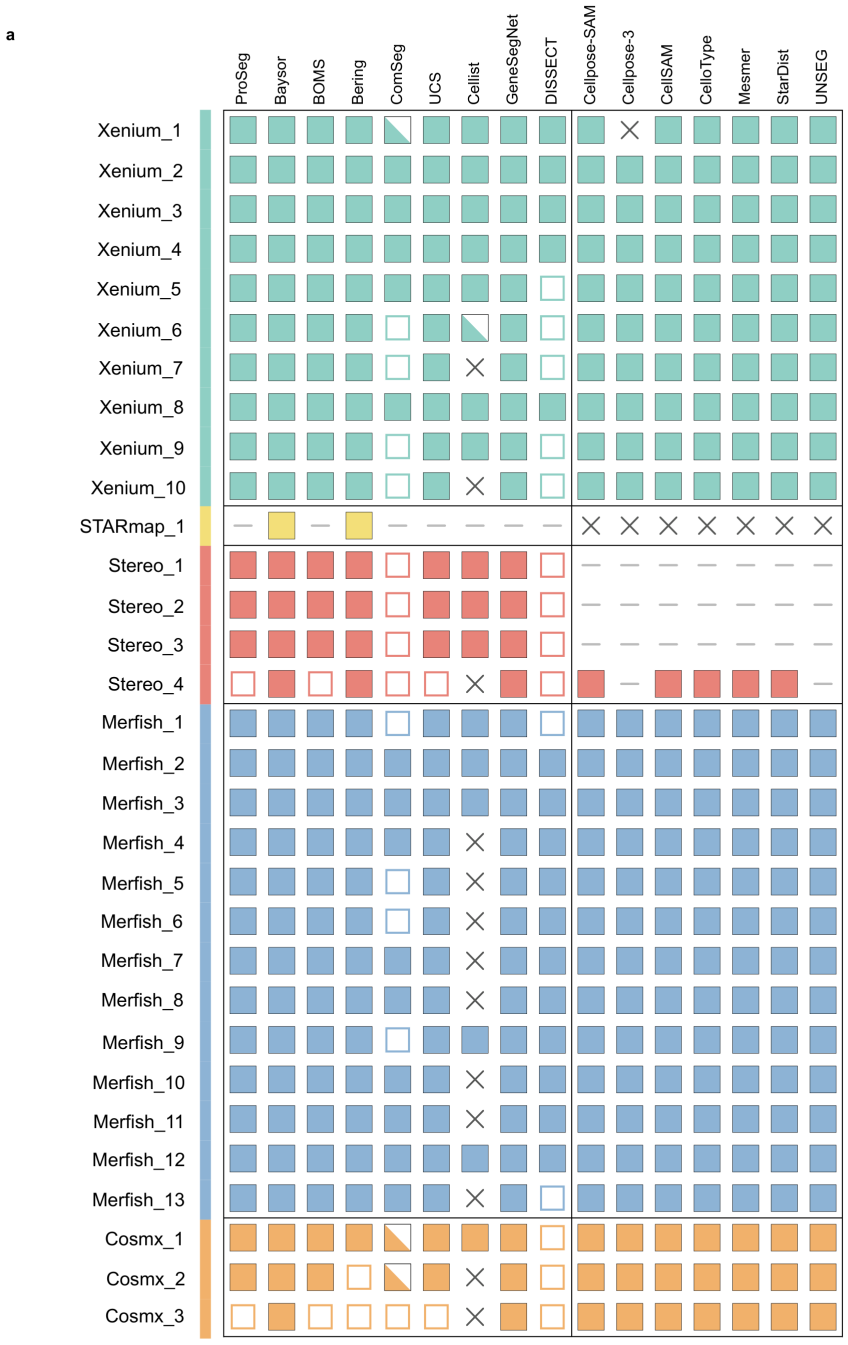

Figure S3. Completion status of all 496 segmentation tasks. Each square is one method run on one dataset. Filled squares mark completed runs, half-filled squares runs on a sub-region, open squares completed runs that did not meet quality-control criteria, crosses failed runs, and grey dashes tasks that were not applicable because the required input was unavailable. Of the 466 applicable tasks, 445 were completed, including 4 on a sub-region and 35 that did not meet quality control, and 21 failed.

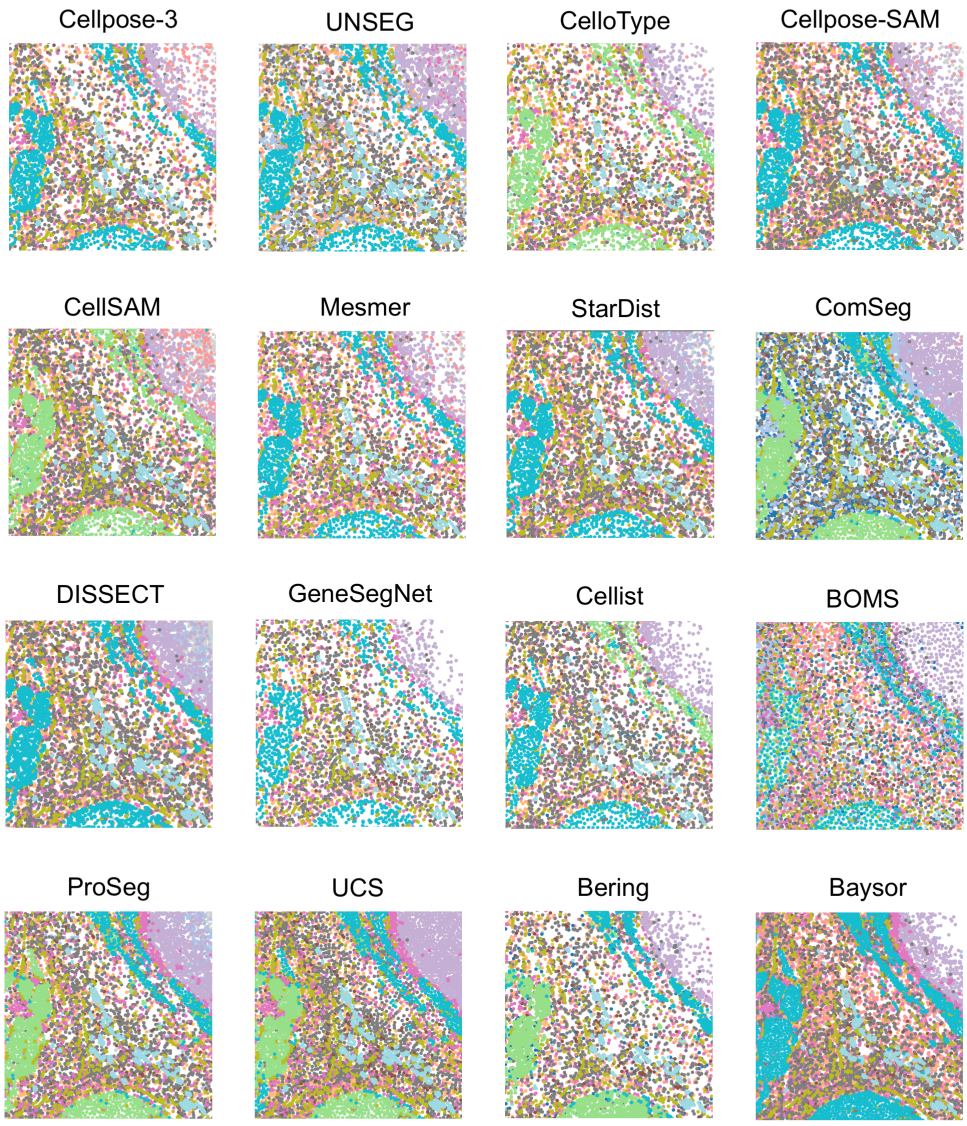

Figure S4. Spatial maps of transcript-based meta-clusters for the segmentation results on MERFISH liver_1 cancer dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

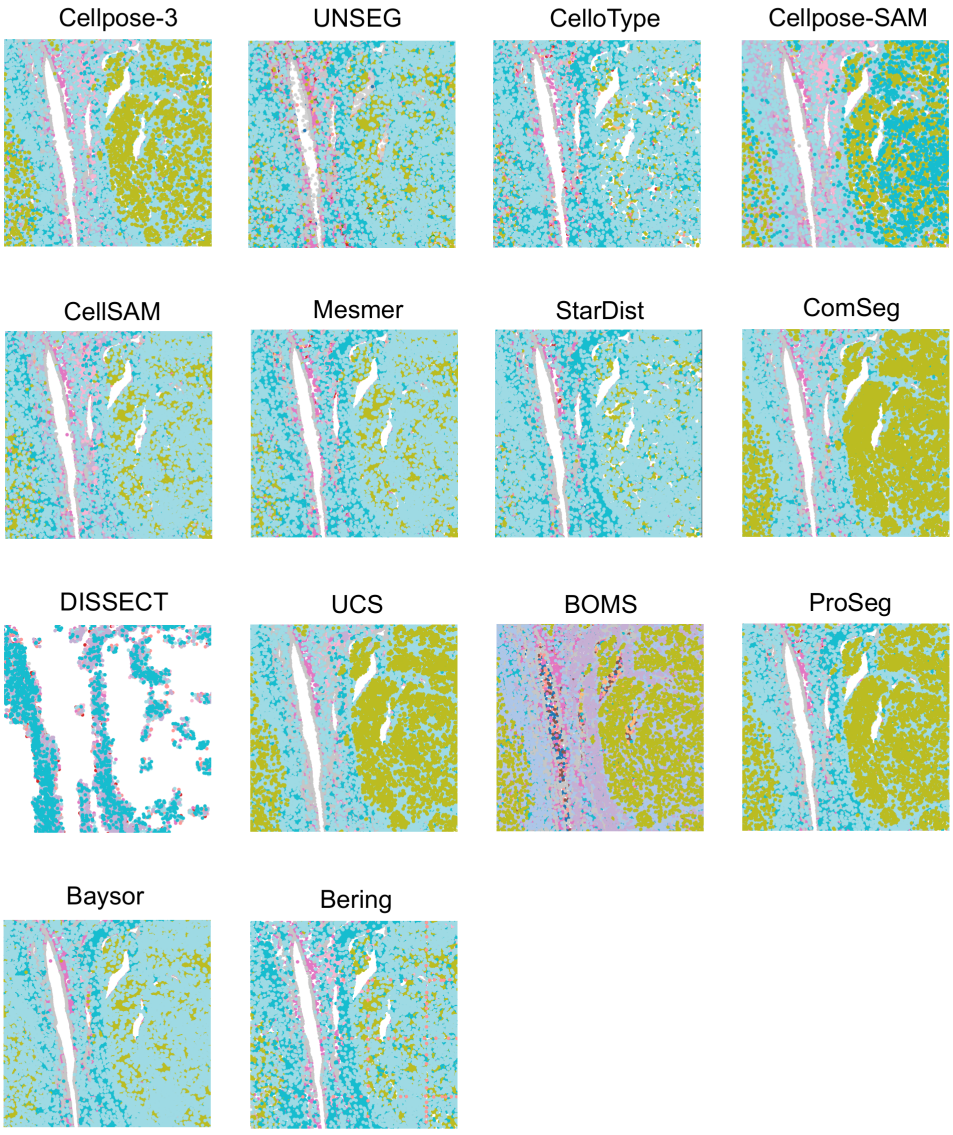

Figure S5. Spatial maps of transcript-based meta-clusters for the segmentation results on Xenium ccRCC dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

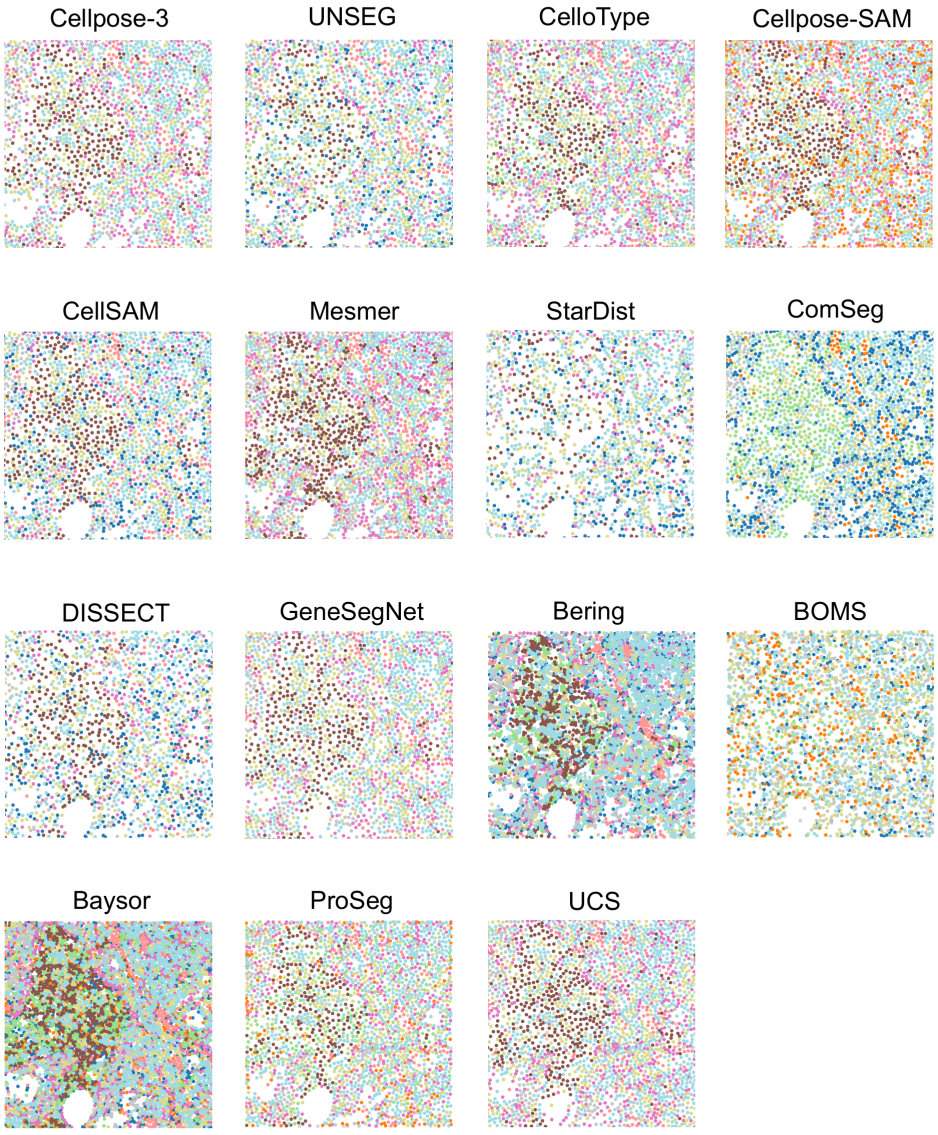

Figure S6. Spatial maps of transcript-based meta-clusters for the segmentation results on Xenium lymph node dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

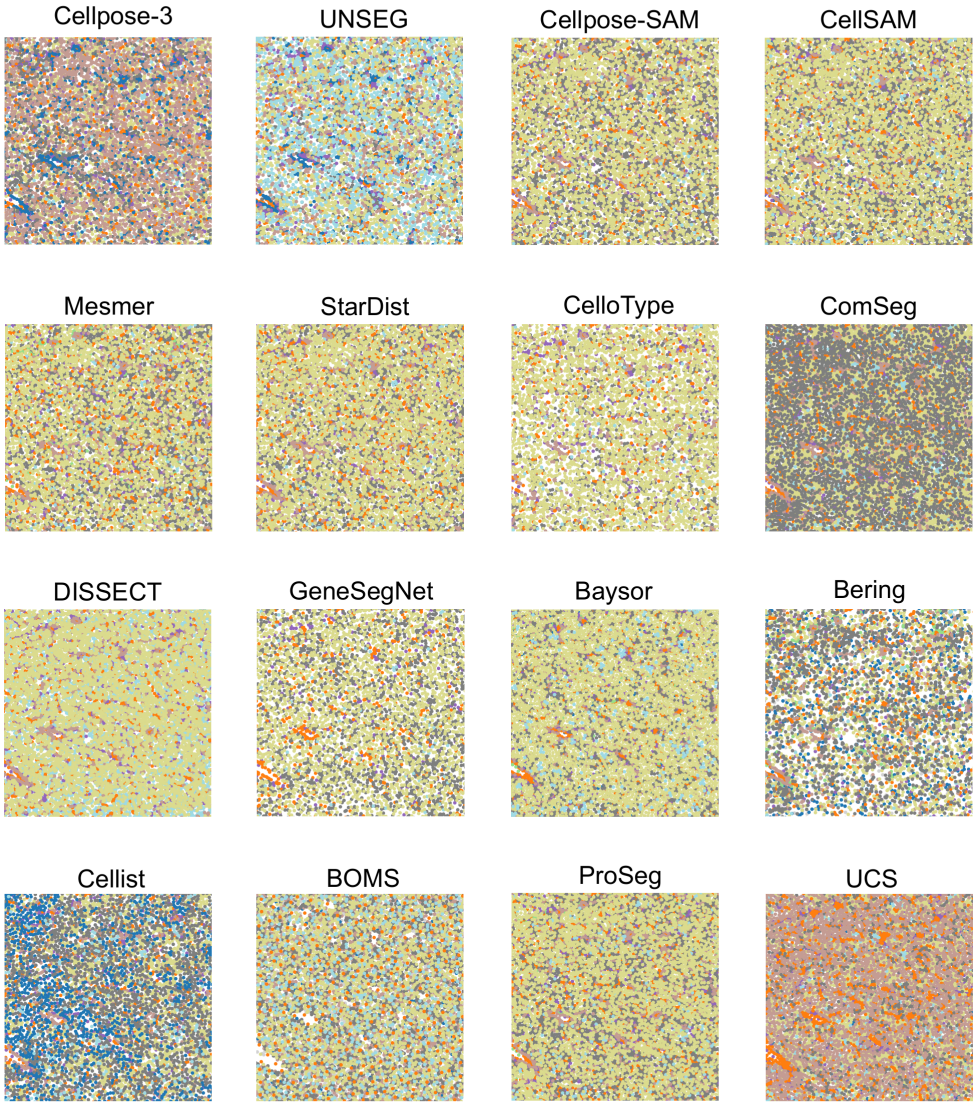

Figure S7. Spatial maps of transcript-based meta-clusters for the segmentation results on MERFISH liver_2 cancer dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

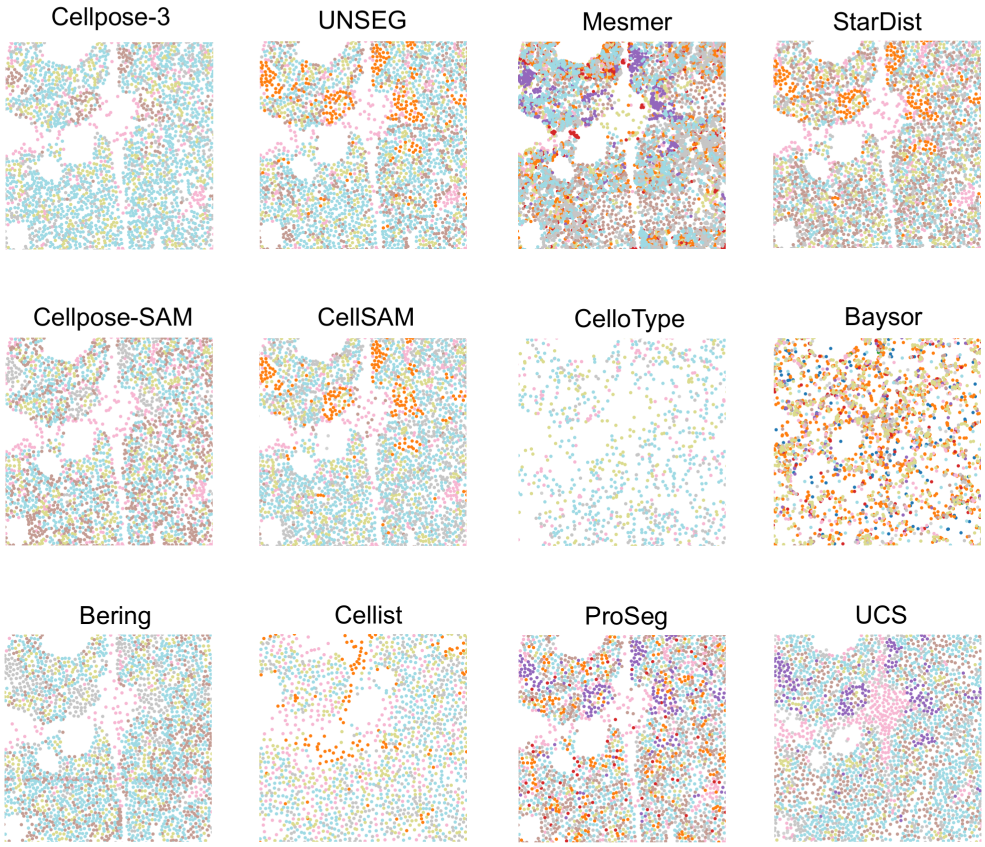

Figure S8. Spatial maps of transcript-based meta-clusters for the segmentation results on CosMx pancreas cancer dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

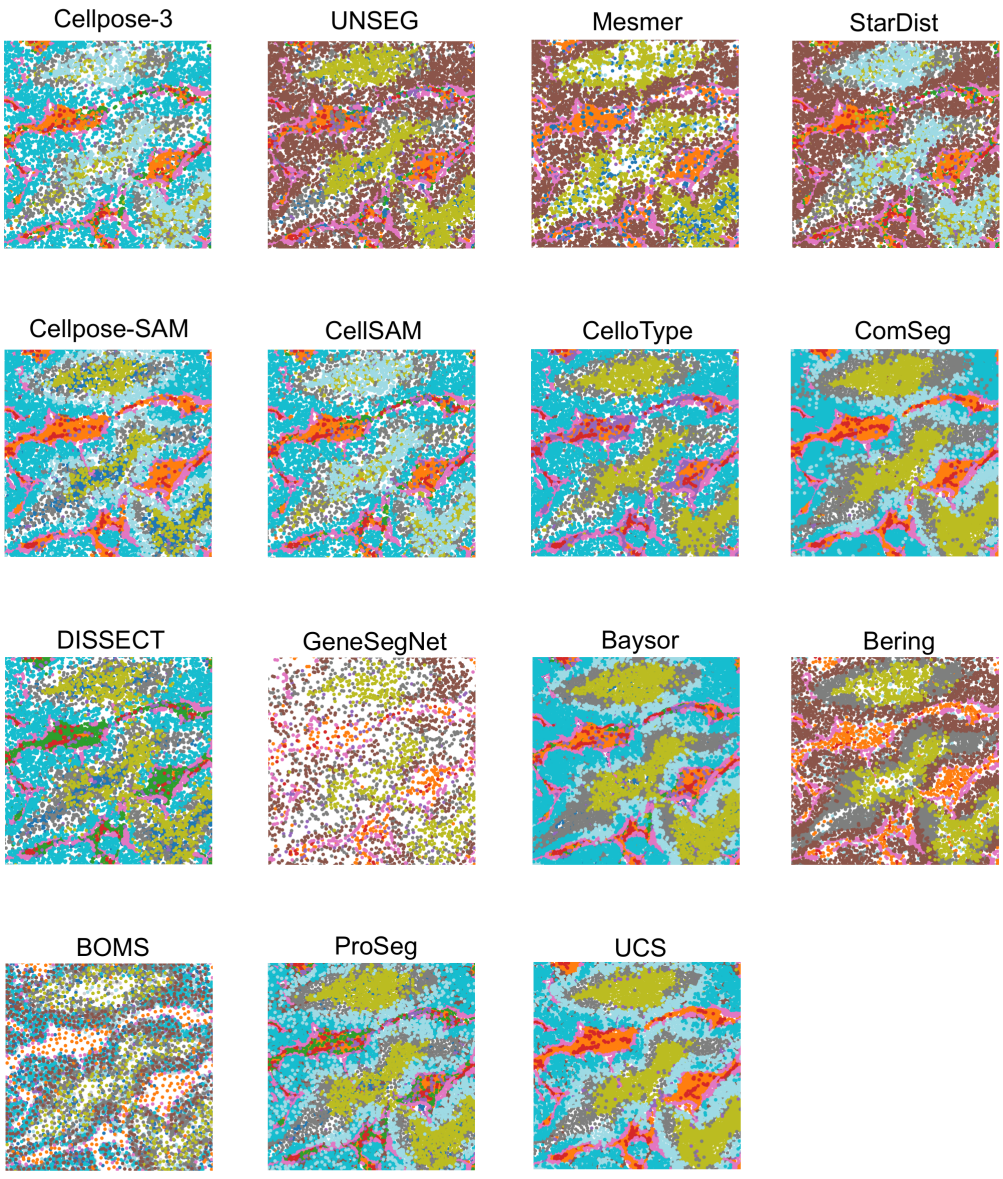

Figure S9. Spatial maps of transcript-based meta-clusters for the segmentation results on MERFISH lung cancer dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

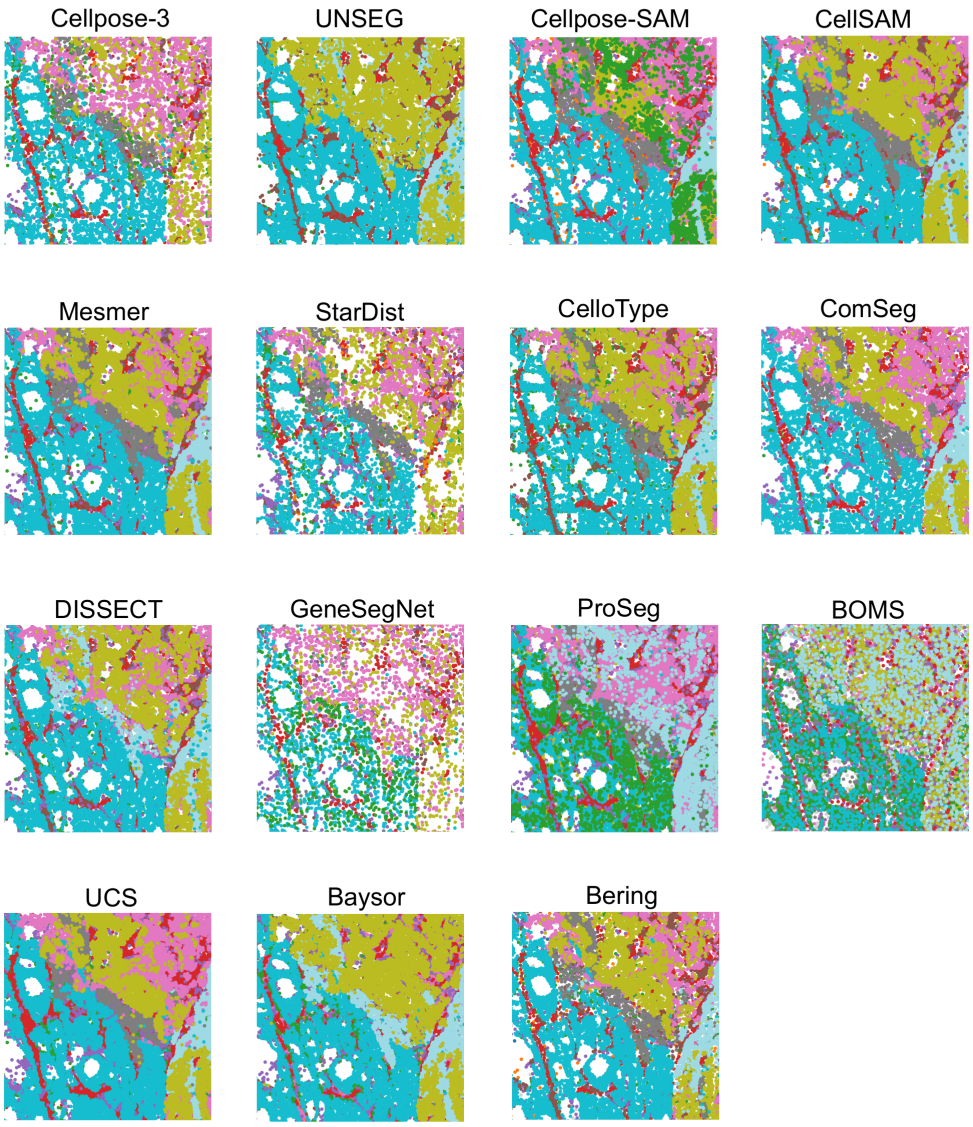

Figure S10. Spatial maps of transcript-based meta-clusters for the segmentation results on MERFISH uterian dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

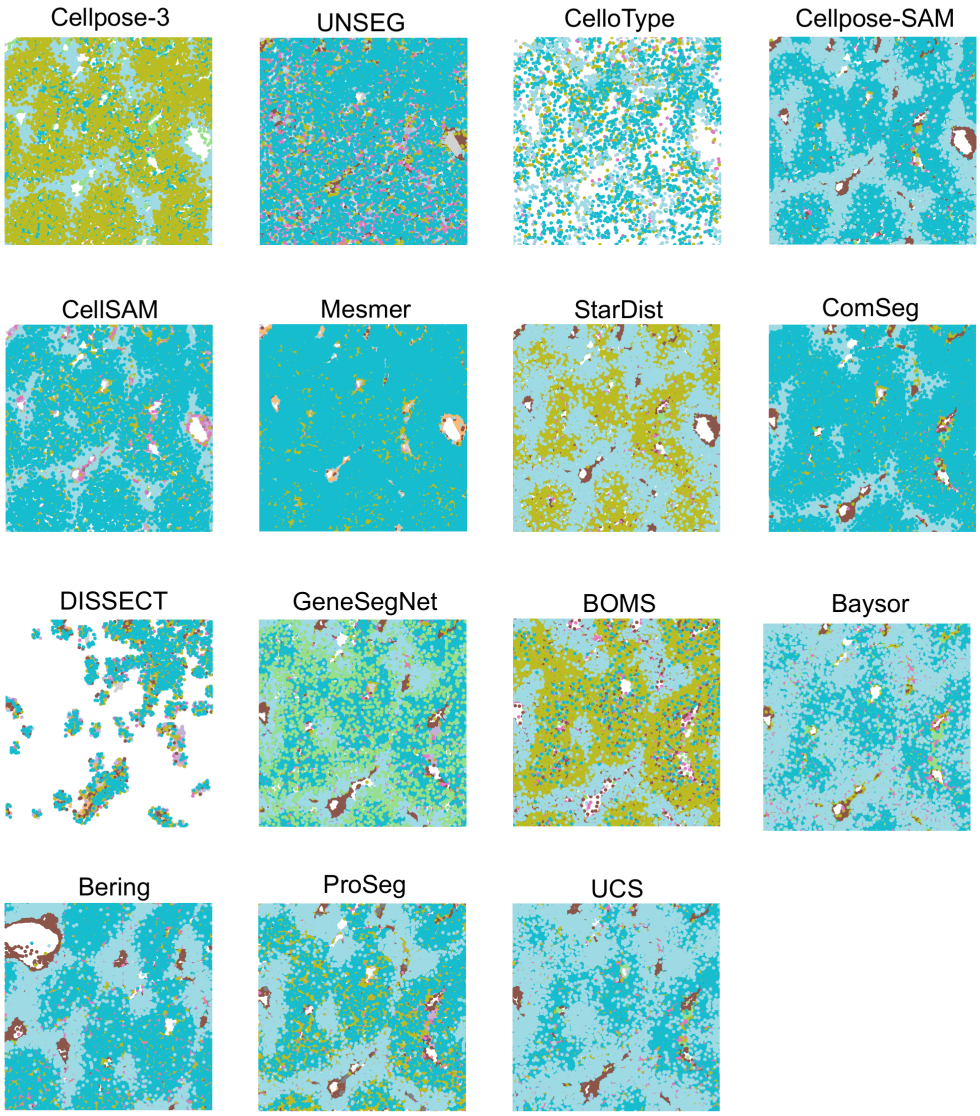

Figure S11. Spatial maps of transcript-based meta-clusters for the segmentation results on Xenium liver dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

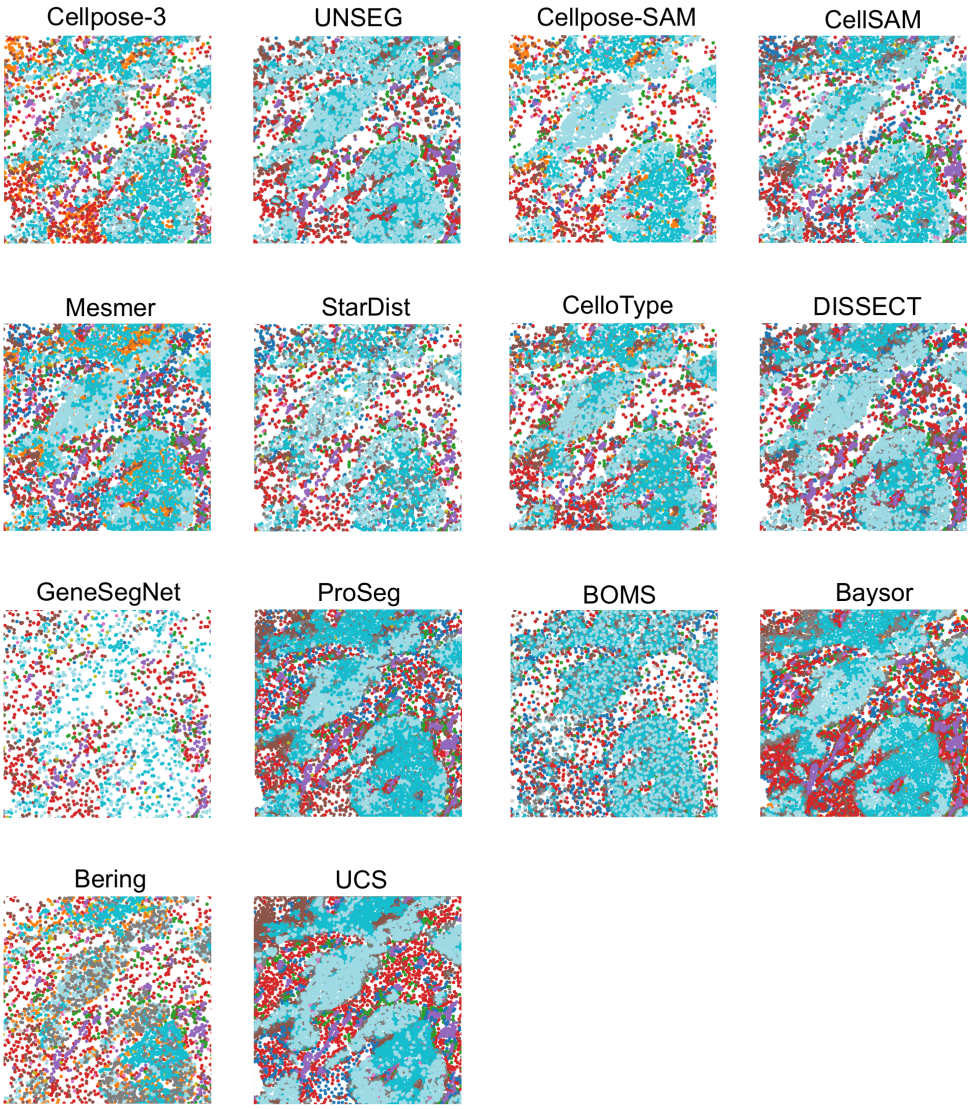

Figure S12. Spatial maps of transcript-based meta-clusters for the segmentation results on MERFISH breast cancer dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

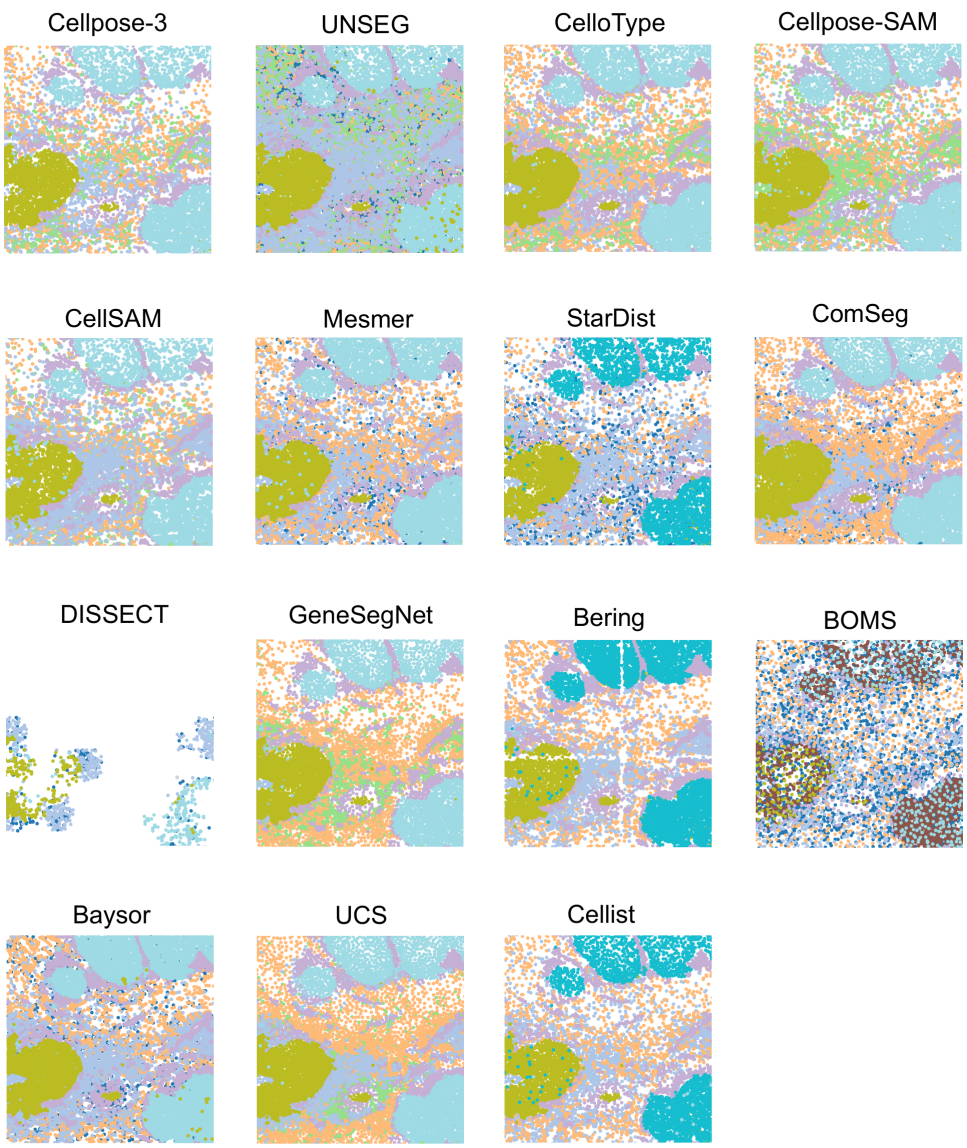

Figure S13. Spatial maps of transcript-based meta-clusters for the segmentation results on Xenium breast dataset. For the segmentation each method, cells were clustered with the Leiden algorithm (resolution 0.6) after normalization, log transformation, and reduction to 30 components. Local clusters were then grouped into dataset-level meta-clusters by hierarchical clustering of their pseudobulk profiles (Pearson correlation of at least 0.65) and drawn at its position.

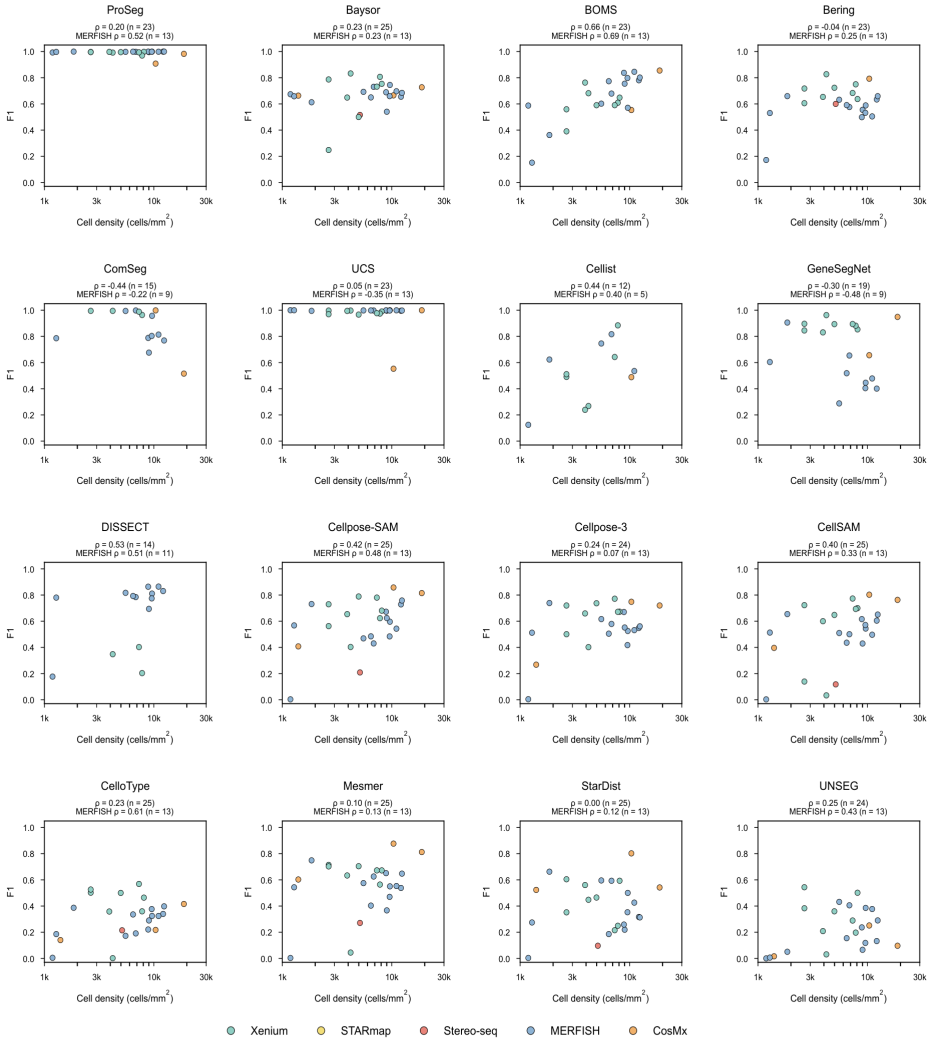

Figure S14. Relationship between cell density and segmentation accuracy for each method. Each panel shows one method plotted by the cell density of its analysis region (reference cells per mm^2, logarithmic scale) and F1, and coloured by platform. Spearman correlations across all datasets and within MERFISH, with the number of datasets (*n*), are given above each panel. F1 is based on mask overlap for image-based models and on centroid matching for transcript-based methods.

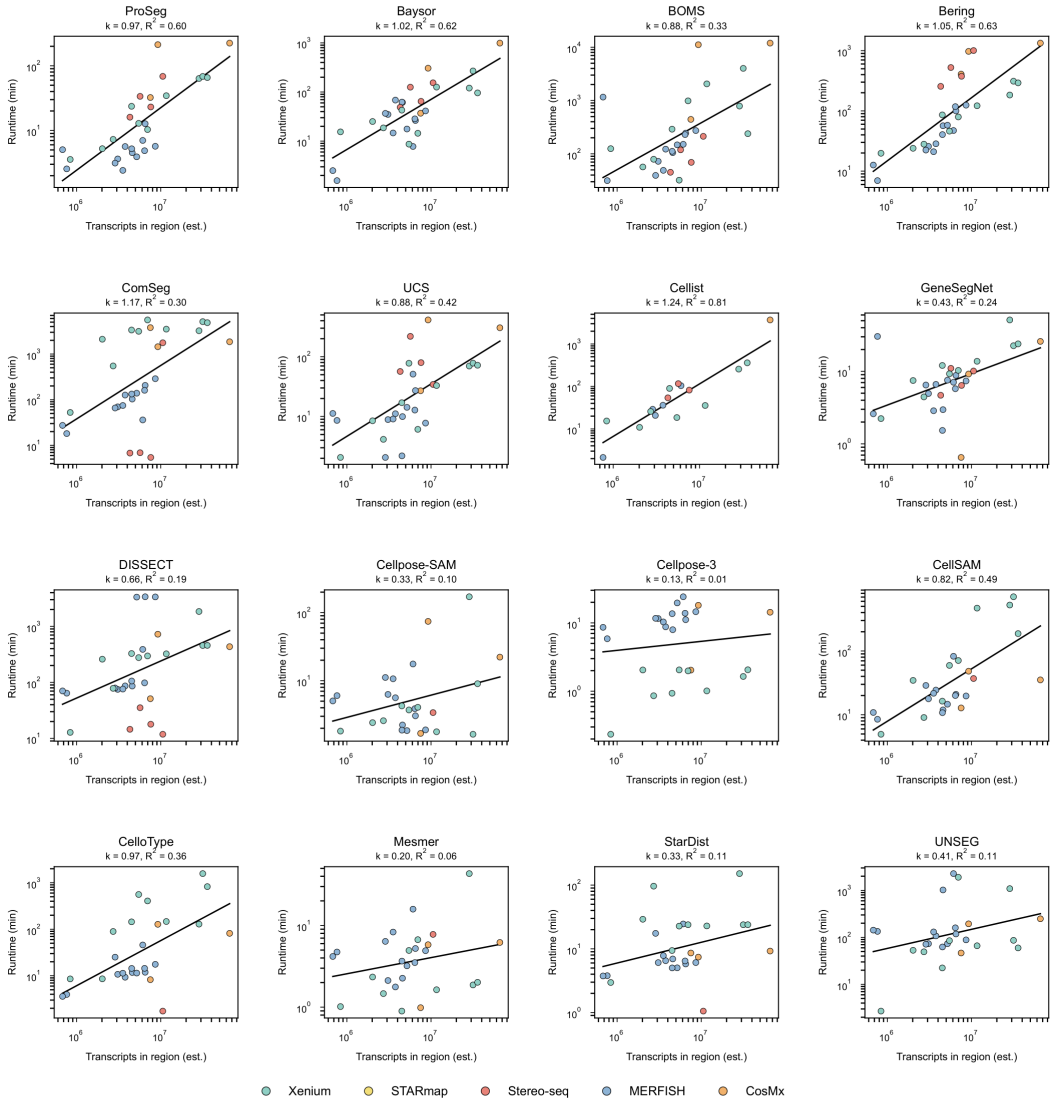

Figure S15. Runtime of each method as a function of the number of transcripts in the analysis region. Each panel shows one method plotted by the number of transcripts in its analysis region and runtime(both on logarithmic scales). Black lines show log-log least-squares fits, with the scaling exponent k and R^2 above each panel, where k close to 1 indicates that runtime grows in proportion to the number of transcripts.

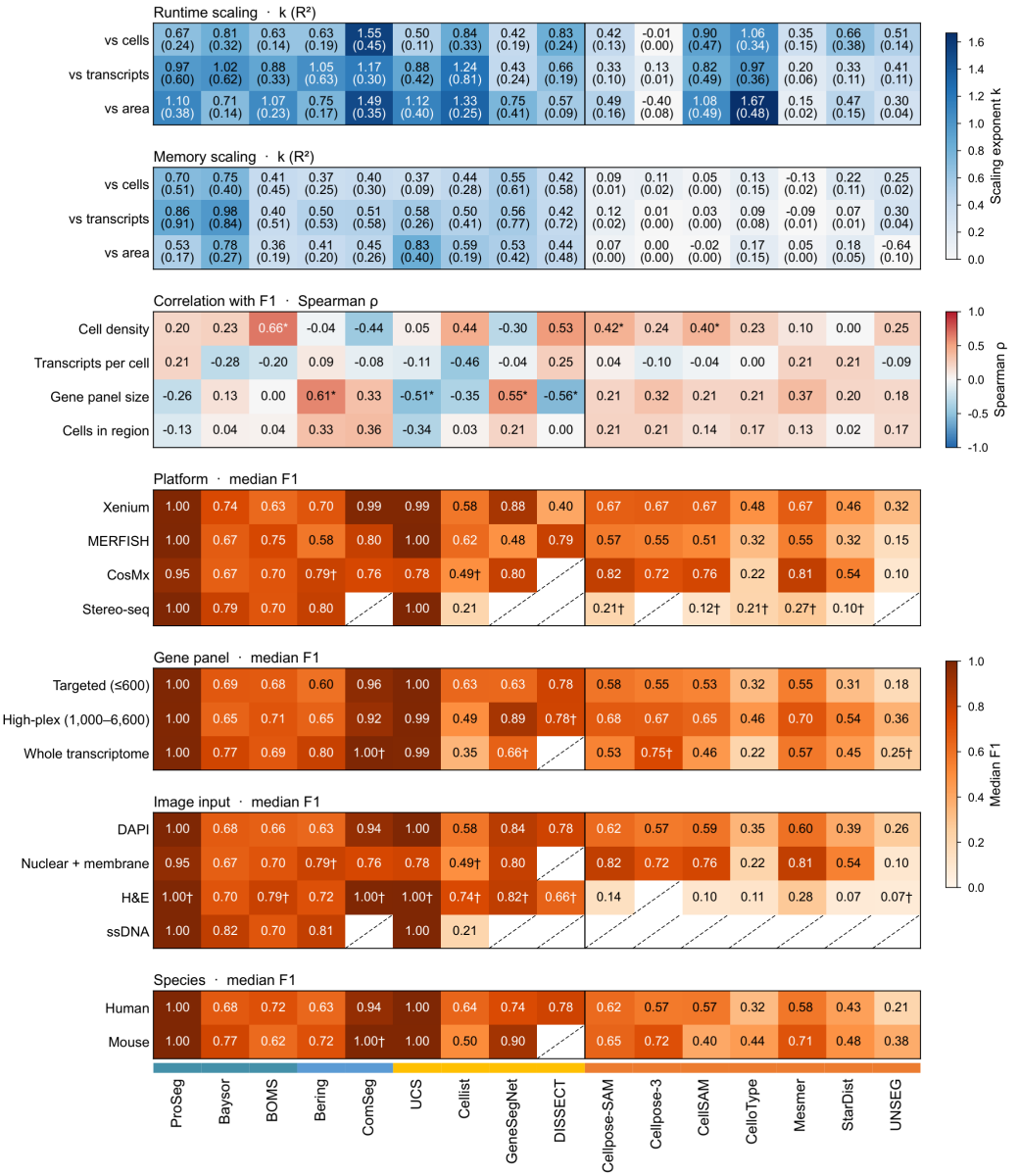

Figure S16. Computational scaling, dependence on dataset characteristics, and stratified accuracy of the 16 methods. **Runtime and memory scaling,** slope k of log–log least-squares fits of runtime or peak CPU memory against the number of cells, the number of transcripts, or the area of the analysis region, with R^2 in parentheses, for methods with at least five datasets (see also Supplementary Fig. S15). **Correlation with F1,** Spearman correlation across datasets between F1 and cell density, transcripts per cell, gene panel size, and cell number, with asterisks marking $P < 0.05$ (see also Supplementary Fig. S14). **Platform, gene panel, image input, and species,** median F1 of each method within each category, with daggers marking single-dataset categories. F1 is based on mask overlap for image-based models and on centroid matching for transcript-based methods.